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Enhancing Business Dashboards with Explanatory Analytics & AI

Exploring the Use of AI and Explanatory Analytics to Enhance Business Decision-Making

Turku School of Economics
Master's thesis

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Abstract

Business dashboards have become increasingly popular for descriptive analytics, significantly aiding the decision-making process. With the growth in data volume and complexity, explanatory models for automatic diagnostic analytics have been developed. This thesis aims to enhance these models, thereby extending the functionality of business dashboards by generating textual explanations instead of relying on traditional visualisations or summarised tables. This approach facilitates more informed decision-making.

This thesis employs design science research, where an artefact is created based on theory. The research evaluates three theoretical frameworks for diagnostic modelling, ultimately incorporating only one: the explanation formalism. By utilising generative AI, the developed artefact translates intricate data insights into easily understandable narratives. This approach bridges the gap between data analysis and decision-making, providing clear and comprehensible explanations of data trends and anomalies.

The created artefact was tested using a cognitive walkthrough, demonstrating its capability to transform complex data figures into comprehensible text format. Additionally, a comparison was made between the outputs of explanation formalism, an explanatory tree, and explanatory text. This thesis contributes to the fields of information management and business intelligence by presenting a method to integrate existing explanatory models with AI. The result is a prototype that enhances the clarity and understanding of business data.

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1 Introduction

This thesis is aimed to research the development of a prototype for extending the use of explanatory analytics with AI as a tool in business dashboards. The goal is to improve business decision making.

1.1 Background

The increasing sophistication of big data technologies and progress in machine learning has generated heightened interest in business intelligence, capturing the focus of both industry professionals and academic researchers (Chen et al., 2012). Notably, progress in business intelligence (BI) and analytics is evident, with more companies incorporating features like explanatory analytics to enhance decision-making. As a result, the significance of business dashboards has risen, serving as a crucial tool to visually present key performance indicators (KPI's) and assist managerial decision-making (Walchshofer et al., 2023). Managers are aided by explanatory analytics to automatically provide the underlying causes.

Although trained managers can swiftly identify the root cause, understanding explanation trees requires training and domain expertise (Eckerson, 2010). Thus, the proposed thesis aims to enhance automated explanatory analytics by integrating AI to simplify the results for users and enhance decision-making.

1.2 Example

The idea is to test AI's usefulness as an explanatory analytics algorithm to be used as a textual explanation for the provided data. Suppose an analyst works for a firm and notices that their firm is doing poorly on return on assets as opposed to the competitors in the industry. The analyst takes a look at the business dashboard and selects the ROA as variable to be explained, the business dashboard computes an explanation which is similar to figure 1; an explanatory tree explaining that the Return On Assets (ROA) is affected by the Profit Margin (PM) and Turnover Assets (TA), indicated by the full line. It also indicates counteracting effects for example the profit margin is negatively affected by the Direct Material Cost (DMC) and the Research and Development Costs (RDC) this is indicated by the dotted line.

While useful for an experienced user of a Business Dashboard it is quite hard for an inexperienced user. Furthermore, even for an experienced user it can take quite some time to understand the tree, especially when the user is new to the organisation or has not worked with the algorithm for some time. Therefore the goal of this thesis is to create an artefact that provides a textual explanation of the underlying reasons. Through making use of AI the end result provides an explanation similar to figure 2.

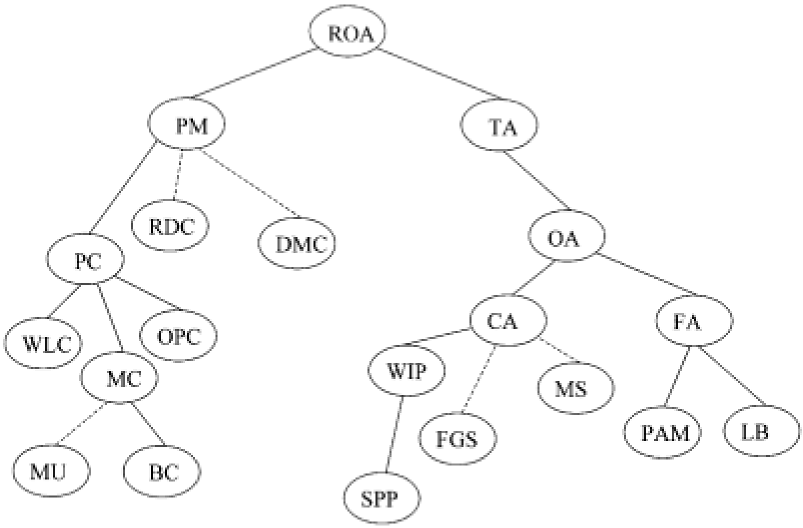


Fig. 2. Diagnosis for $S = \{\partial\text{ROA} = \text{low}\}$ at firm 9.

Figure 1: Explanatory Tree for inter-firm comparison(Feelders and Daniels, 2001)

Conclusion on Factors Affecting ROA for Firm 9

The analysis of Firm 9's financial metrics highlights several key factors contributing to its lower Return on Assets (ROA) compared to other firms in the industry:

1. **High Production Costs**: Firm 9 has substantial production costs (PC = 0.774), which is a major factor dragging down the Profit Margin (PM = 0.089). These costs are composed of Material Costs (MC), Wage and Labor Costs (WLC), Waste/Spoilage (Ws), and Other Production Costs (OPC). Each component's relatively high value suggests inefficiencies in production processes and potentially excessive labor costs.
2. **Inefficient Asset Utilization**: Firm 9's Total Assets (TA = 0.9) are not utilized efficiently, as indicated by the low Operating Assets (OA = 0.99). This inefficiency is further evidenced by the composition of Current Assets (CA) and Fixed Assets (FA), where Work in Process (WIP = 0.225) is particularly high, suggesting delays in production or inventory issues.
3. **Cost Structure Impact on Profitability**: Direct Material Costs (DMC = 0.046) and Administrative Costs (AC = 0.091) directly reduce PM. While DMC might indicate issues with sourcing or procurement, high AC points towards possible overhead inefficiencies.
4. **Asset Composition Challenges**: The asset structure, comprising Current Assets (CA) and Fixed Assets (FA), shows a significant portion of the assets in non-liquid forms such as WIP, which affects cash flow and operational flexibility.

These elements collectively lead to Firm 9's lower ROA, underscoring issues in production efficiency, asset management, and cost control, which are critical areas affecting the firm's financial health and operational effectiveness.

Figure 2: Goal of this thesis

1.3 Research Question

The main research question of this thesis is:

"How can the integration of artificial intelligence (AI) techniques enhance the explanatory analytics capabilities of business dashboards to improve business decision making?"

The supporting questions i.e. sub-questions are found below:

1. What are business dashboards, explanatory analytics, prompt engineering and what do they entail?
2. What are the requirements for an artefact that uses AI as a tool for explanatory analytics?
3. How to develop an AI-powered artefact for explanatory analytics using Python?
4. How to evaluate the addition of AI to explanatory analytics within a business dashboard environment?

1.4 Research Relevance

1.4.1 Business Relevance

The incorporation of Artificial Intelligence (AI) into the existing explanatory analytics framework within business dashboards introduces several business implications. This integration enhances the depth and sophistication of data interpretation, providing nuanced insights and contextualised explanations.

AI contributes to a deeper understanding of complex patterns and trends within textual data, offering a more detailed perspective on observed phenomena. This enhances the interpretability of analytics results, enabling stakeholders to comprehend specific circumstances and contributing factors behind outcomes or trends.

Moreover, the use of AI's in explanatory analytics supports advanced decision-making by providing richer and more detailed insights. Decision-makers can make informed choices based on a thorough understanding of the underlying data, contributing to contextually aware decision support.

In addition, AI's play a role in efficient problem resolution, offering a simple explanation of the root causes of issues. This can expedite the decision-making process and facilitate the timely resolution of challenges identified through the analytics process.

Furthermore, the integration of AI's enhances communication by generating human-readable explanations. This improvement in communication makes it easier for stakeholders to comprehend and act upon the insights presented in business dashboards, fostering a more collaborative and informed decision-making environment.

1.4.2 Scientific Relevance

The scientific relevance is aimed at adding knowledge to existing research of explanatory analytics as well as research in business intelligence in general. The development in the field of data-analysing system are becoming more transparent, but tend to be complex for a layman to use (Montavon et al., 2018). Secondly by using AI to support explanatory analytics this research hopes to achieve a more granular perspective on observed phenomena. This heightened granularity contributes to the interpret ability of analytics results and provides a contributing factor behind outcomes or trends. By comparing explanatory analytics to explanatory analytics with AI, certain limitations or advantages are discovered.

1.5 Research Methodology

For this thesis the main research methodology is design research. The goal is to create an artefact that provide a quick text summary of visuals,tables and reports. To develop this artefact the model by Hevner et al. (2004) is used for the creation of this artefact. This model has three components: the environment, the artefact and the knowledge base. The following figure shows an adapted version of the design science framework from Hevner et al. (2004), which is used in this thesis.

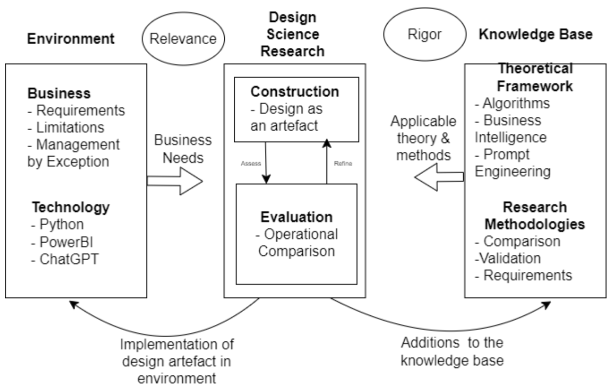


Figure 3: Adapted Design Science Research Framework Hevner et al. (2004)

The knowledge base consists of two components namely the theoretical framework and research methodologies. figure 3 shows that the theoretical framework is made up of algorithms, prompt engineering and business intelligence. During a literature review concepts such as business dashboards, AI and prompt engineering used to answer Research Question 1: What are business dashboards, explanatory analytics, prompt engineering and what do they entail?

Based on the foundation of the theoretical knowledge the requirements and use cases are established. Furthermore, again based on the theoretical knowledge the type of technologies that are used in this thesis are discussed. This knowledge answers Research Question 2: What are the requirements for an artefact that uses AI as a tool for explanatory analytics?

To proceed with the design phase, the next crucial step involves the creating of the artefact.

This process entails leveraging artificial intelligence (AI) technologies. The synthesis of these components, in conjunction with AI, aims to enhance the functionality and effectiveness of the resulting artefact. The artefact is coded in Python and harnesses the power of GPT-4 realise its objectives and answer Research Question 3: How to develop an AI-powered artefact for explanatory analytics using Python?

To validate the efficacy and robustness of the developed artefact, comprehensive testing procedures are essential. Initially, the artefact undergoes rigorous testing against the predefined requirements outlined in chapter 3. This phase ensures that the artefact aligns with the specified objectives and functions as intended. Furthermore, to evaluate the artefact's usability and effectiveness in, it is subjected to an operational comparison with PowerBI to assess various facets of its performance in order to answer research question 4 How to evaluate the addition of AI to explanatory analytics within a business dashboard environment?

1.6 Thesis Scope

This thesis spans various research domains, bringing together intersecting areas of study. The general goal of this thesis is to create an artefact that translates the found causes by explanatory analytics into text. This generated text provides offers enhanced comprehension of underlying causes and is therefore part of diagnostics analytics. The process of reasoning enhances the expertise of business analysts and enables them with superior insights. opportunities for decision making. Therefore it also falls within the area of Business Intelligence.

A limitation of this study is that it is a prototype and not fully functioning software that can be evaluated. This thesis makes use of Python and therefore the artefact may not work for other programming languages.

1.7 Thesis Outline

This thesis follows a structured format where sub-question one is answered in chapter 2, sub-question two is in answered in chapter 3 etc. Following the introductory chapter, Chapter 2 delves into a review of existing literature related to business dashboards. Moving on to Chapter 3, it presents a theoretical framework for explanatory analytics, offering an overview of various models used in business dashboards. Chapter 4 expands on explanatory analytics by incorporating AI and sets the requirements for the artefact of this thesis.

Chapter 5 explains the creation process of the artefact, outlining its components. Chapter 6 focuses on evaluating the artefact. Finally, Chapter 7 provides a summary along with a discussion, conclusion, and considerations of limitations.

2 Navigating the Explanatory Analytics Landscape

In this section of the thesis, the emphasis is on exploring the theory behind several key concepts, including business dashboards, explanatory analytics, and prompt engineering, as well as their related components. To fully comprehend business dashboards, it is essential to first grasp the concept of business intelligence, since business dashboards originate from this broader field.

Moreover, within this chapter, the explanatory analytics models are discussed, thereby laying the necessary groundwork for subsequent implementation phases. Through a comprehensive examination, a clearer understanding of their applicability and implications within the context of our study is provided.

Furthermore, this chapter delves into the concept of prompt engineering, shedding light on the pivotal factors to ensure the efficient and accurate attainment of intended objectives.

By traversing through these theoretical domains, a foundation is established with the requisite knowledge and insights necessary to inform the subsequent phases of the research, thereby laying a robust foundation for the practical implementation and evaluation of our proposed methodologies and frameworks.

2.1 Business Intelligence and Business Dashboards

As mentioned in the previous paragraph, this sub chapter undertakes a thorough exploration into the theoretical foundations of business dashboards and their integral relationship within the broader framework of business intelligence (BI). The goal is to provide an in-depth analysis of business dashboards which also highlights their connection to the overarching field of BI.

Within this sub chapter, a comprehensive examination of business dashboards, spanning their historical evolution, conceptual frameworks, and functional attributes. Furthermore, the analysis extends to scrutinising the pivotal role of business intelligence as the conceptual precursor and enabling framework for business dashboards.

2.1.1 Business Intelligence

The evolution of the business intelligence (BI) field has been characterised by transitioning from the use of structured internal data to incorporating unstructured external data, ultimately integrating mobile and sensor-based content, as observed by Chen et al. (2012). This evolution hasn't occurred in isolation; rather, it's been accompanied by a parallel advancement in technologies and analytics, striving to accommodate the bigger volume and heterogeneity of data sources flooding the modern business landscape (Antunes et al., 2022). In the context of this thesis, the focus rests on structured data sourced from a dashboard platform, where data cleansing has already been conducted, streamlining the process for analysis and decision-making (Negash, 2004). figure 4 below gives an example of a dashboard which provides a visual representation of how structured data is organised, visualised, and utilised for strategic decision-making within an organisation. By presenting data in a clear and accessible manner, dashboards facilitate quick insights and informed decisions.

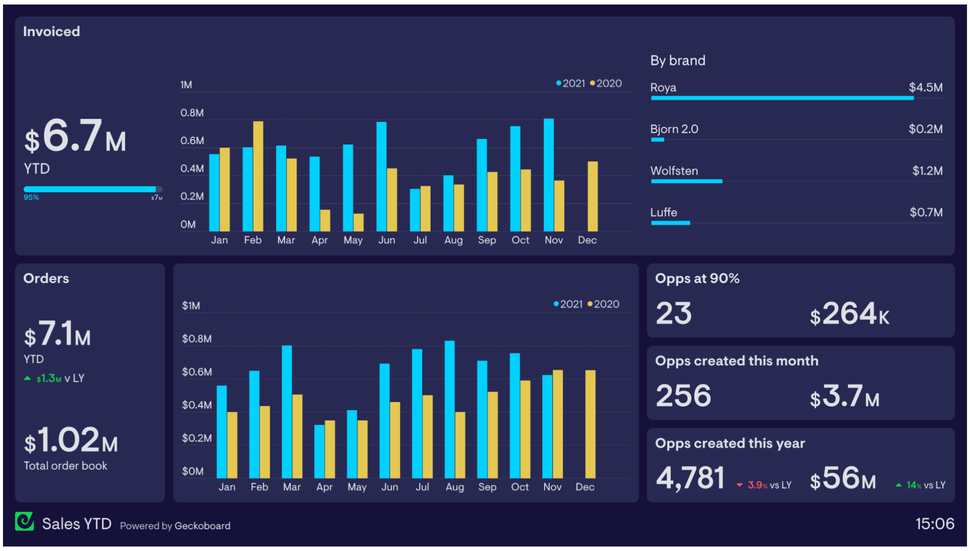


Figure 4: Example of a sales business dashboard(Geckoboard, 2024)

To understand the essence of business intelligence (BI) and its evolution, a glance at the diverse definitions that have emerged over time provides valuable insight. Božič and Dimovski (2019) emphasise the multifaceted nature of BI, highlighting its pivotal role in transforming raw data into actionable insights that underpin strategic decision-making within organisations. Notably, business intelligence has been conceptualised as a product as well as a process, as articulated by Jourdan et al. (2008), where the product aids organisations in predicting their operating environment to inform strategic decisions effectively. Moreover, technology emerges as a cornerstone in defining BI, as underscored by Joshi et al. (2010) and Bose (2009), reflecting the relationship between technological advancements and the evolution of BI.Chen et al. (2012) provide a comprehensive definition of BI, encapsulating a number of methods, devices, architectures, customs, approaches, and uses aimed at empowering organisations in their decision-making endeavours. In alignment with the objectives of this thesis, BI is conceptualised as a multifaceted entity encompassing processes, technologies, and products that facilitate informed decision-making within organisations, drawing upon automated diagnostic processes, robust technological platforms, and tangible outcomes derived from the thesis artefact.

The fundamental value proposition of BI lies in its ability to serve as a catalyst for informed decision-making within organisations, enabling invaluable insights through BI products. However, determining the value creation within BI unveils a complex overview of interrelated factors, as illuminated by Božič and Dimovski (2019). Hence, subsequent subsections delve deeper into the process, product, technology, and value creation facets of BI. Golfarelli et al. (2004) say the process of business intelligence is a holistic event transcending mere data warehousing, encompassing the transformation of data into actionable intelligence across operational, tactical, and strategic levels. This synthesis of the ETL(Extract, Transfer, Load) process and data warehousing, coupled with business rules, leads to a user-friendly interfaces that empower strategic decision-making. The BI architecture figure 5 illustrates the layered structure of BI systems, showing data sources, ETL processes, data warehouses, and analytical tools.

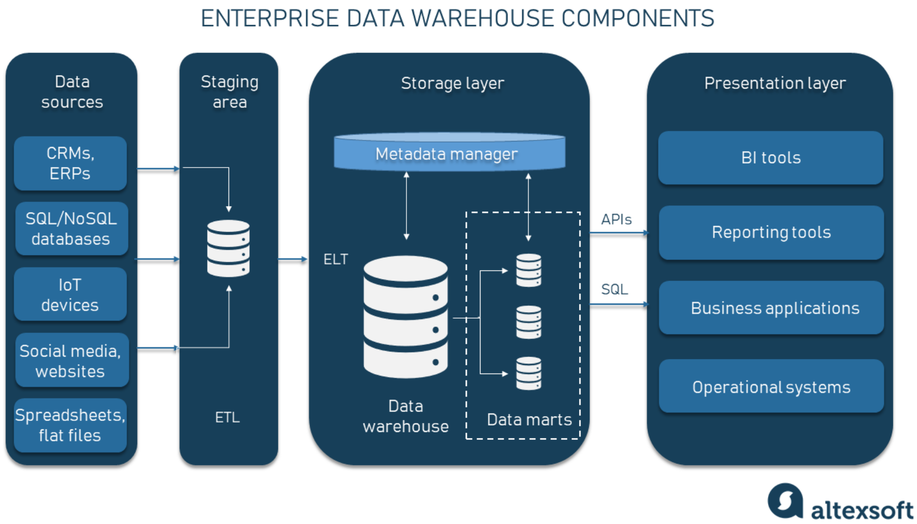


Figure 5: BI architecture (Altexsoft, 2023)

The criticality of contextualising data within business knowledge is emphasised by March and Hevner (2007), who advocate for robust infrastructure and procedural frameworks to leverage resultant insights effectively. Empirical findings by Isik et al. (2012) underscore the indispensable role of technological capabilities and data quality in fortifying BI's support for organisational decision-making, drawing attention to the relationship between adept BI processes and investments in technological infrastructure.

This idea is furthered by improved methods for handling unstructured data. From an analytical standpoint, there is a focus on understanding quantitative data using statistical and mathematical models.There are different types of data analysis, dividing them into descriptive, predictive, and prescriptive analyses and diagnostic analysis.

- Descriptive analysis addresses the questions of 'what?' and 'why?' regarding past events, commonly manifested through reports, data visualisations, and dashboards.
- Predictive analysis forecasts the probability of future events by utilizing techniques like data mining, machine learning, and statistical analysis to uncover patterns within the data.
- Prescriptive analysis pertains to optimisation, forecasting the outcomes of potential actions to determine the most favourable approach.
- Diagnostic analysis centres on elucidating the causes behind observed outcomes, seeking to answer the question "why did it happen?" by uncovering underlying factors.

In alignment with the thesis's focal point, which centres on internal and structured data, the BI definition hones in on harnessing the potential of these data types, leveraging the ETL process to streamline data utilisation. Moreover, technological prowess emerges as an important driver in enhancing the presentation, visualisation, speed, and accuracy of BI products.

To further clarify this thesis focal point figure 6 is depicted. It shows the essential process of this thesis. Namely the interaction between the business database and multidimensional model which feed the visualisation in the business dashboard. When the business decision maker notices a variable which he/she wants explained the explanatory analytics generate an explanation, in this case an explanatory tree, which the business decision maker can use for making decisions.

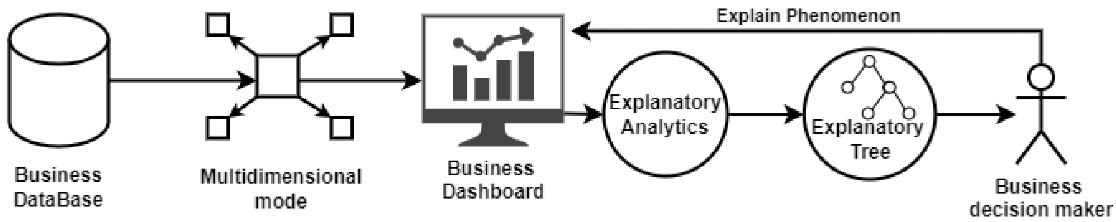


Figure 6: Business Intelligence focus of this thesis

In conclusion, the evolution of business intelligence (BI) has seen a significant transformation from reliance on internal and structured data to the integration of external and unstructured data sources, along with the adoption of mobile and sensor-based content. This evolution, has occurred with advancements in technology and analytics, driven by the necessity to accommodate the increasing volume and diversity of data flooding modern business environments. Central to the understanding of BI is its multifaceted nature, encapsulating both process and product dimensions. In essence the ability of BI lies in its ability to empower organisations with informed decision-making capabilities, facilitated by invaluable insights derived from BI products.

2.1.2 Data Modelling

Dimensional modelling is a design technique used in data warehousing to structure data for easy access and high performance in query processing. It organises data into fact tables and dimension tables, with the fact tables containing measurable, quantitative data about business processes, and the dimension tables holding descriptive attributes related to the facts. This method simplifies complex data schemas, making it intuitive for users to retrieve information and perform analyses. By creating a clear separation between numeric data (facts) and descriptive context (dimensions), dimensional modelling supports efficient and insightful business intelligence operations. This architecture offers a consistent and intuitive structure that enables efficient data access (Kimball and Ross, 2011).

The star schema, a common approach in dimensional modelling, includes a central fact table connected to multiple dimension tables. The fact table captures metrics from business process events, while the dimension tables offer contextual information related to these metrics (Tryfona et al., 1999). Typically, each fact table is linked to the dimension tables in a one-to-many relationship, where each dimension table includes columns that provide various levels of detail. These often form hierarchical structures, enabling data analysis at different levels of aggregation, such as month, quarter, and year.

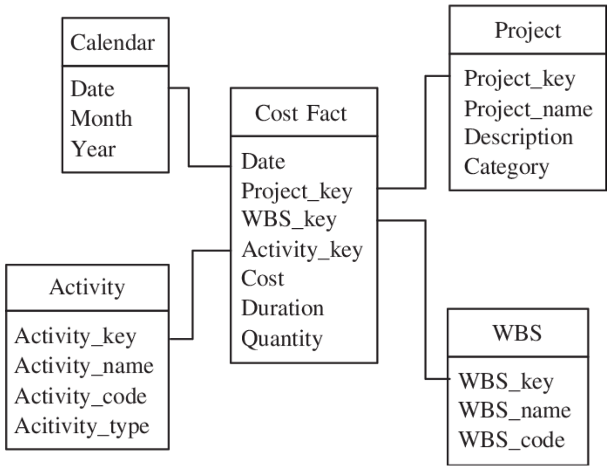


Figure 7: a Star model with one central table and four dimension tables. (Rujirayanyong and Shi, 2006)

The Snowflake Relational Model, represents an evolution of the traditional relational database model. This model is designed to address several limitations found in conventional relational databases, particularly regarding performance and scalability. One of the key innovations in this model is the separation of storage and compute resources. By decoupling these two components, Snowflake allows for independent scaling of storage and computational power. This means users can increase computational resources to handle heavy queries without the need to expand storage capacity, and vice versa, leading to significant improvements in efficiency and cost management.(Levene and Loizou, 2003)

Additionally, the Snowflake Relational Model is built with the cloud in mind, leveraging cloud infrastructure to offer automatic scaling capabilities. This automatic scaling ensures that the computational resources adjust dynamically based on the workload, providing optimal performance and cost efficiency at all times. Another significant advantage of this model is its support for semi-structured data, such as JSON and XML. Snowflake integrates and processes this type of data, offering flexibility that traditional relational databases often lack (Dageville et al., 2016)

Online Analytical Processing (OLAP) technology is crucial for BI. OLAP cubes and databases hold aggregated data in multidimensional formats, allowing for the organisation and analysis of large datasets. This technology greatly improves query speed because of the pre-computed aggregated data. OLAP cubes enable complex queries across dimensions such as Time Period, Product Type, and Location for sales information. OLAP operations support various data navigation and manipulation tasks, typically illustrated as operations on a data cube

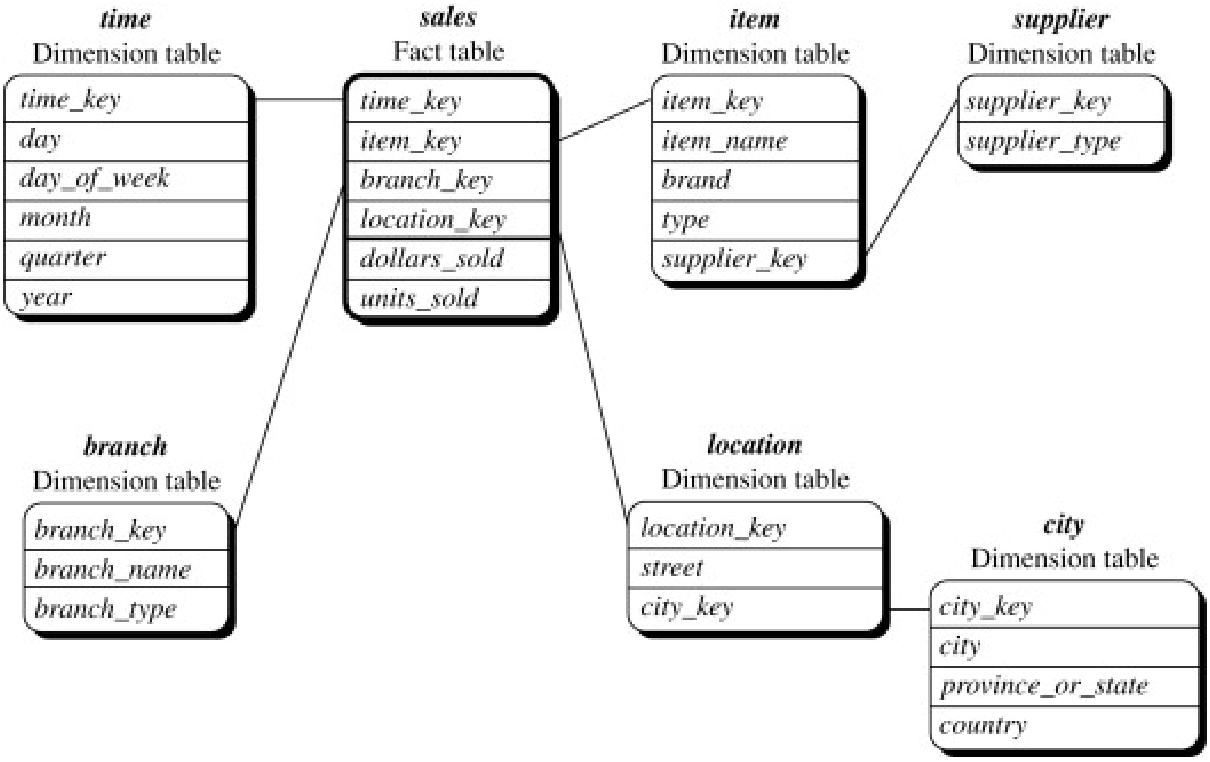


Figure 8: Snowflake model (Tupper, 2011)

(Chaudhuri and Dayal, 1997).

OLAP cubes are designed to address complex queries and data analysis tasks efficiently. They organise data into multiple dimensions, allowing users to analyse information across various perspectives. For instance, a sales data cube might include dimensions such as time (year, quarter, month), geography (country, region, store), and product categories. This structure enables detailed and comparative analyses quickly.

One of the core advantages of OLAP cubes is the storage of pre-aggregated data. Common calculations, such as totals, averages, and counts, are pre-computed and stored within the cube. As a result, queries can be executed rapidly because the system does not need to compute these aggregations on the fly. This pre-aggregation significantly enhances query speed, which is particularly beneficial for real-time analysis and decision-making processes.

OLAP cubes support a range of complex querying operations, essential for thorough data analysis. Roll-up operations aggregate data along a dimension, moving up the hierarchy (e.g., from months to quarters). Drill-down operations break down data into finer granularity (e.g., from years to quarters). The slice operation extracts a single layer of data from the cube, similar to taking a "slice" out of it (e.g., sales data for a particular year). Dice operations select a sub-cube by specifying multiple dimensions (e.g., sales data for a particular year and region). The pivot operation rotates the cube to view data from different perspectives, helping discover insights from different angles.

The differentiation between dimensional databases and OLAP is crucial. While OLAP

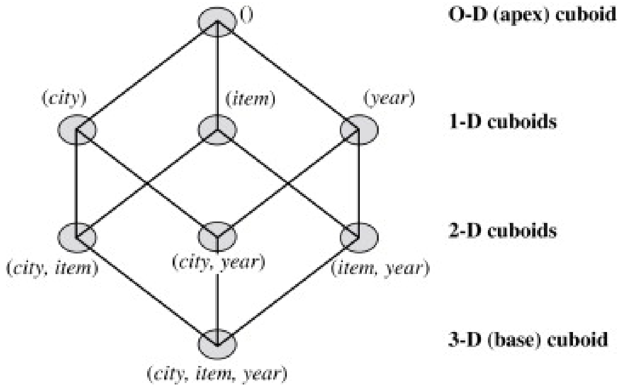


Figure 9: Olap Data Cube (Han et al., 2011)

uses a multidimensional structure to store pre-aggregated data, dimensional databases use flat, decentralised tables: fact tables and dimension tables. The two databases leverage identical structural principles from the data warehouse's multi-dimensional schema, underscoring the importance in researching explanatory OLAP analytics.

Dimensional databases organise data into fact tables, which contain quantitative data, and dimension tables, which describe dimensions such as time, geography, and product. They follow a star or snowflake schema and are optimised for query performance and data retrieval. In contrast, OLAP systems build on the dimensional model by creating cubes that allow for fast, complex queries and analysis. They go beyond the capabilities of dimensional databases by pre-aggregating data, which enhances performance and enables sophisticated analytical operations.

2.1.3 Business Dashboards

Business dashboards serve as the keystone in modern data analysis frameworks, seamlessly connecting data processing pipelines with end-user comprehension and decision-making (Tanko et al., 2019). Harnessing the power of data visualisation, these dashboards transform complex datasets into digestible insights, facilitating informed decision-making processes across various organisational levels (Rasmussen et al., 2009).

The significance of visualising data cannot be overstated. The ability of visualisations to uncover meaningful insights and relationships within vast datasets is crucial. By simplifying information retrieval and reporting, dashboards provide decision-makers with intuitive tools to navigate through the complexity of data analysis, empowering them to extract actionable insights efficiently (Chaudhuri et al., 2011).

Dashboards serve a multifaceted purpose within organisations, acting as the bridge between decision-makers and Business Intelligence and Analytics systems. (Foley and Guillemette, 2010). They offer a comprehensive view of key performance indicators (KPIs), enabling stakeholders to monitor organisational performance in real-time and identify areas that require attention or intervention (Eckerson, 2010). Additionally, dashboards facilitate trend analysis, anomaly detection, and pattern recognition, equipping professionals with the tools they need to make data-driven decisions swiftly and effectively (Khatuwal and Puri, 2022).

While the utility of dashboards is undeniable, one of their primary limitations lies in their inability to provide automated explanations for observed phenomena (Pauliuk, 2018). For instance, while a dashboard may highlight a decrease in sales for a specific district, it may not offer insights into the underlying causes of this decline. To address this challenge, there is a growing emphasis on integrating visualised algorithmic models into dashboard designs, enabling users to gain deeper insights into the factors driving observed trends and anomalies.

Integrating business dashboards into an organisation's Business Intelligence architecture typically involves leveraging data warehousing solutions (Thusoo et al., 2010). Data warehousing involves gathering, consolidating, and structuring data from various sources to support decision-making and generate business insights. Dashboards are positioned within the visualisation layer of this architecture, serving as the front-end interface for accessing and interacting with data warehouses (Rasmussen et al., 2009).

At the heart of a business dashboard lies its back-end infrastructure, which often relies on dimensional databases and star schema to organise and store data (Rasmussen et al., 2009). The star schema enables multidimensional analysis, allowing users to explore data from various perspectives and dimensions (Eckerson, 2010). On the front-end, dashboards offer a range of applications tailored to different user needs, including monitoring, analysis, and management (Eckerson, 2010). These applications ensure that dashboards align with organisational objectives and facilitate decision-making across all levels of the enterprise.

In conclusion, business dashboards serve as indispensable tools for navigating the complexities of modern data analysis. By leveraging the power of visualisation and integrating with robust BI Analytics systems, these dashboards empower decision-makers to extract actionable insights from vast datasets, driving organisational success and innovation.

2.2 Managerial Decision Making

Business dashboards are instrumental tools employed by business analysts for generating explanations. Analysts utilise these explanations to investigate anomalies they have observed. For instance, an analyst might notice that store X reports lower revenue compared to other stores, despite similar conditions across all locations. In such scenarios, a business dashboard enables the analyst to delve deeper into the data, identifying root causes for the anomaly and facilitating informed corrective actions (Negash and Gray, 2008).

While manual analysis is possible, dashboards offer real-time overviews of relevant information, significantly expediting the process. This immediacy allows analysts to swiftly uncover underlying issues and make timely decisions. Consequently, dashboards enhance the efficiency and effectiveness of the analytical process, supporting better business outcomes (Negash and Gray, 2008).

In short a business dashboard supports the decision making process and is a decision support system. Mintzberg (1989) describes the decision-making process not as linear but as consisting of multiple phases and various actions, which he developed into a comprehensive model. His model consists of three phases. The Identification Phase starts with the recognition of a problem or opportunity, followed by diagnosing the situation to understand its root causes and implications. In the Development Phase, decision-makers search for potential solutions or alternatives and then design these solutions in detail. Finally, in the Selection Phase, the alternatives are screened to narrow down the options, evaluated and compared in detail, and the final decision is made and authorised for implementation. A detailed depiction of the model is found below.

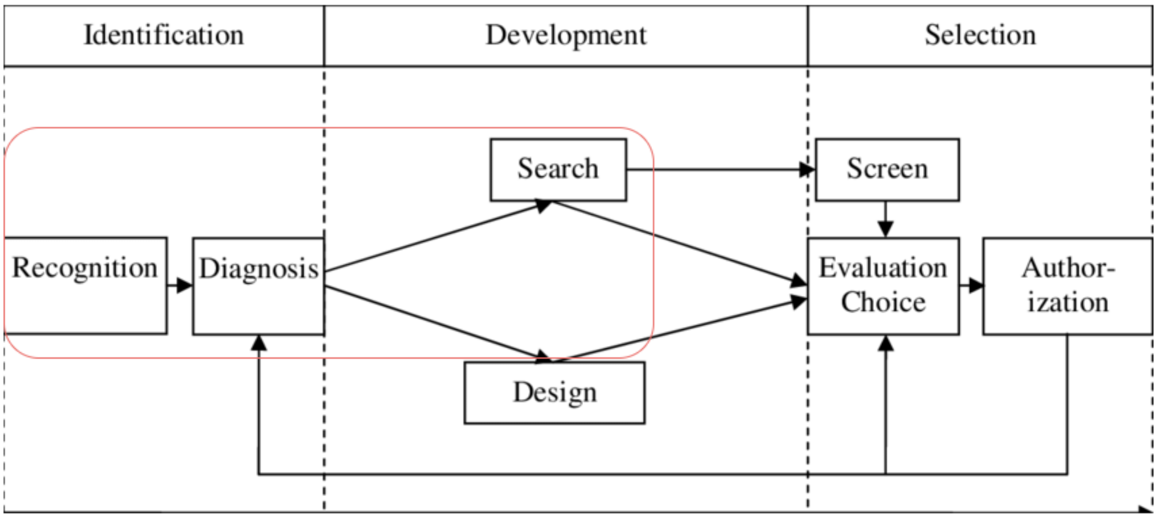


Figure 10: Model of decision making process(Kask, 2019)

In this thesis context, the recognition, diagnosis and search aspects - which are highlighted in the figure above - of Mintzberg's decision-making model are particularly essential. These phases lay the groundwork for informed and effective decision-making by ensuring that the problem or opportunity is thoroughly understood before moving on to potential solutions.

The process begins with the recognition phase. This phase is initiated when a business analyst observes something unusual or noteworthy on the business dashboard. This could be an unexpected dip in sales, a sudden spike in customer complaints, or any other anomaly that deviates from the norm.

Once an anomaly is recognised, the process moves into the diagnosis phase. During this phase, explanatory analytics are employed to delve deeper into the data and uncover the root causes of the observed anomaly. This involves a thorough analysis of various factors that could be contributing to the issue. For instance, if there is a sudden drop in sales, the analyst might explore data related to market trends, competitor actions, internal operational issues, or changes in customer behaviour. Advanced analytical techniques, such as regression analysis, hypothesis testing, and data mining, are often utilised to identify correlations and causation's.

During the development phase, solutions are shaped to address the problem. The search aspect within this phase involves looking for existing solutions within the organisation's repertoire. This can include consulting past decisions or examining formal procedures that were previously successful. The objective is to find a proven solution that can be adapted to the current situation efficiently.

2.3 Explanatory Analytics and Dashboards

Explanatory analytics, or Explanation Generation, seeks to identify the underlying causes of observed events, making it a subset of diagnostic analytics. Causal explanations are valuable because they clarify the factors influencing an event, thus highlighting the relevance and significance of the explanation (Friedman, 1974).

Explanatory analytics is a data analysis methodology aimed at understanding causal relationships within datasets. It explains why certain phenomena occur by using techniques from statistics, econometrics, and machine learning, providing a rigorous framework for establishing causation, essential for reliable decisions and interventions.

This approach addresses the critical need to understand causation behind data patterns, rather than merely identifying correlations. In a business context, this helps uncover the reasons for various outcomes and performance metrics, preventing misguided decisions based on superficial insights.

For example, a retail company experiencing fluctuating sales might find that descriptive analytics reveal sales peaks in December and drops in January, while predictive analytics forecasts similar trends. However, neither explains the fluctuations. Explanatory analytics investigates potential causes such as marketing campaigns, economic conditions, competitor actions, seasonal behaviours, or consumer preferences. This helps the company determine that December sales spikes are driven by holiday promotions and increased consumer spending, while January drops result from post-holiday fatigue and lower marketing efforts, allowing for targeted strategies to mitigate the slump (Davazdahemami et al., 2022).

Despite its benefits, explanatory analytics presents challenges that can hinder its effectiveness. Sophisticated statistical methods and machine learning techniques can be daunting for individuals unfamiliar with these concepts, creating a steep learning curve requiring extensive training (Moubayed et al., 2018). Even for those comfortable with the technical aspects, lacking domain expertise can be a significant barrier. Accurate interpretation and effective decision-making require a deep understanding of the data's specific context and nuances (Viaene, 2013).

For individuals who haven't engaged with data analytics for a while, reacquainting themselves with the technique can be difficult due to the rapid evolution of analytical tools and methodologies. This can make previously learned techniques obsolete or require significant relearning.

3 Explanatory Models in Business Dashboards

This section dives into algorithms that are used in business dashboards. First, the current explanatory analytics models are explained, providing an in-depth look at the methodologies and frameworks that help interpret complex data and generate actionable insights. Following this, the section describes previous applications of these models, showcasing real-world examples and case studies that highlight their effectiveness and versatility across different industries.

The next subsection focuses on generative AI and prompt engineering, exploring the advanced techniques and tools that enable machines to create content, generate responses, and perform tasks traditionally done by humans. This part covers the principles of prompt engineering, its significance in enhancing AI capabilities, and practical examples of its implementation.

Following the discussion on generative AI, the section includes a comprehensive comparison between explanatory analytics models, previous applications, and generative AI. This comparative analysis highlights the strengths, weaknesses, and unique aspects of each approach, providing a clear understanding of how they complement or differ from one another.

Lastly, the current applications of AI in business dashboards are evaluated, examining how these advanced technologies are being integrated into modern business intelligence tools to improve decision-making, efficiency, and overall performance.

The next subsections; 3.1, 3.2 and 3.3 are about the current explanatory models. The first subsection begins with a detailed explanation of the formalism algorithm, delving into the methodology and processes involved in the formalisation of explanations. The following subsection explores informative summarization. The last section is about the explanation by intervention algorithm.

3.1 Explanation Formalism

Explanation formalism is a structured approach to generating causal explanations for observed events in multidimensional databases. It is based on a format which Feelders and Daniels (2001) designed and which was later extended by (Caron, 2013). It clarifies specific facts by drawing comparisons between objects using a real value (F), which outlines the comparison, and a reference value (r). The reference values (r) are determined by a normative model. This method is particularly useful in OLAP (Online Analytical Processing) systems and business dashboards, where the goal is to diagnose and explain deviations from expected values. Typical comparisons include those with historical data and industry benchmarks. These values can be derived through various methods, depending on the relevant models. Financial planning, sales performance tracking, operational efficiency assessments, and customer behaviour analysis models rely on data straight from the data cube itself. Explanations are defined as:

$$\langle a, F, r \rangle \text{ because } C_b, \text{ despite } C_a.$$

Here, $\langle a, F, r \rangle$ represents the event to be explained, where a is the actual value, F is the property showing deviation from the reference object r. C_b is the collection of contributing causes, and C_a is the collection of counteracting causes. For a causal explanation, C_b must contain at least one element, whereas C_a can be empty. This model simplifies in diagnostic contexts to:

$$\delta y = q \text{ occurred because } C_b, \text{ despite } C_a.$$

.

Feelders and Daniels (2001) assert that an explanation relies upon fundamental principles that describe relationships such as causality or interdependence's among variables. These relationships can be depicted through a business model M. It represents relevant financial and operating variables. Using equations such as $y = f(x)$, where $x = (x_1, x_2, x_3, \ldots, x_n)$. A business model M can be visualised as an explanatory graph, showing dependencies between variables.

In this model, variables on the left side of the equation are influenced by or depend on variables on the right side. This representation clarifies how changes in certain variables impact others within the system. For example, in a financial context, revenue might depend on units sold and price per unit, while costs depend on fixed and variable expenses. Quantitative explanations use these equations to measure the impact of one variable on another, aiding in decision-making by predicting outcomes of changes in key variables.

For qualitative differences with values like {low, normal, high}, a measure of influence must be defined to assess the impact of the contributing as well as the counteracting causes. Involved are the quantifying effects of changes in qualitative variables, such as how different levels of customer satisfaction influence repeat purchases or brand loyalty.

The model also accounts for interactions between variables, which can sometimes have non-linear effects. For instance, improving product quality might significantly boost customer satisfaction if the current quality is low but have less impact if the quality is already high. By considering these nuances, the business model M provides a framework for analysing both quantitative and qualitative factors, facilitating better strategic planning and problem-solving.

3.1.1 Measure of influence

The measure of influence ($\text{inf}(x_i, y)$) represents the difference between the actual value and the normal value of y when only x_i deviates from the norm. This adheres to the *ceteris paribus* principle, assessing the impact of x_i on y while holding all other variables constant. The influence measure for an individual variable is defined as:

$$\text{inf}(x_i, y) = f(\mathbf{x}^r_{-i}, x_i^a) - y^r.$$

It signifies the difference between the actual value and the normal value of y if only x_i differs from the norm, thereby adhering to the *ceteris paribus* principle. This implies that the analysis assumes all other variables remain unchanged while x_i varies. The x_i's are regarded as the causes of y. This metric enables the operationalisation of the concepts of contributing and counteracting causes. For a set of variables X with the index set $I_x \subseteq \{1, 2, \ldots, n\}$ on y, the influence measure is defined as:

$$\inf(X, y) = f(x_r^{-I_X}, x_a^{I_X}) - y_r.$$

This measure provides a quantitative way to evaluate the impact of individual variables within a system. By isolating the effect of a single variable x_i while holding others constant, it becomes possible to determine the extent to which x_i influences y. This approach is particularly useful in complex systems where multiple factors interact, allowing analysts to identify key drivers and inhibitors of outcomes.

Furthermore, this measure can be applied to various fields such as economics, medicine, and engineering, where understanding the individual contributions of variables is crucial for decision-making and strategy development. By quantifying the influence of each variable, stakeholders can prioritise actions and interventions that are most likely to yield desired results.

Where $f(x_r^{-I_X}, x_a^{I_X})$ computes a value by assessing all components in I_X at their actual values while keeping the remaining components of x at their standard values. The direction of the influence measure remains consistent and unaffected by the context. If a variable's influence has the same sign as δy, it signifies that the variable is a contributing cause. On the other hand, if the influence measure's sign is opposite to δy, the variable serves as a counteracting cause.

Contributing causes (C_b) are those where the measure of influence is positive, meaning they drive the deviation of y. Counteracting causes (C_a) have a negative measure of influence, mitigating the deviation. The parsimonious set of causes (C^p) includes only the most significant causes, those exceeding a certain threshold of influence. Contributing parsimonious and counteracting causes require a certain influence on y to exceed a fraction T^+. In this fraction T^+ has an chosen number between 0 and 1 determined by the analyst:

$$\frac{\inf(C_p^+, y)}{\inf(C^+, y)} \geq T^+.$$

The influence measure requires the fulfilment of two distinct constraints: the consistency constraint and the conjunctives constraint. The consistency constraint specifies that both reference values and actual values must adhere to the same functional requirements, i.e., $y_a = f(x_a)$ and $y_r = f(x_r)$. The conjunctives constraint applies to a quantitative equation and holds when, for all subsets $X \subset \{x_i, \ldots, x_n\}/\{x_i\}$:

$$\inf(x_i, y) \geq 0 \implies \inf(X \cup \{x_i\}, y) \geq \inf(X, y).$$

$$\inf(x_i, y) \leq 0 \implies \inf(X \cup \{x_i\}, y) \leq \inf(X, y).$$

Fundamentally, this constraint indicates that the direction of influence of an individual variable should remain stable when considered alongside other variables. This means that if a specific variable x_i exerts a positive influence, it should continue to exert a positive influence even when other variables are added into the equation. Such consistency is essential for combining significant causes into a coherent and unified set. By ensuring that the direction of influence does not change, the model remains interpretable and reliable, which is crucial for accurate analysis and effective decision-making.

When examining business issues, it is important to highlight that monotonic and additive functions are the two main types of functions that satisfy the conjunctive constraint. Monotonicity, in this scenario, refers to the independent monotonic behaviour of each variable, where the relationship between variables moves consistently in one direction—either increasing or decreasing. Nearly all financial models exhibit monotonicity, indicating that this constraint is generally met in most financial contexts.

Monotonic functions ensure that as one variable changes, its influence on the outcome remains predictable. This characteristic is crucial in financial models, where understanding the impact of each variable on financial outcomes is essential for decision-making. Additive functions combine the effects of multiple variables in a straightforward manner, with the overall influence being the sum of individual influences.

The prevalence of these two types of functions—monotonic and additive—in financial models highlights their utility in business analysis. Their adherence to the conjunctive constraint ensures that analysts and decision-makers can rely on these models for consistent and reliable insights, which is vital for effective business problem analysis and constructing accurate financial models.

3.1.2 Maximal Explanation

A one-level explanation identifies the most basic causes of $\delta y = q$. In explanatory analytics, it's essential to find an explanation that thoroughly accounts for δy. Achieving a comprehensive explanation for an observed difference involves repeatedly identifying the set of contributing parsimonious causes. This process starts by pinpointing the contributing and counteracting causes, then selecting the simplest set among them. If a variable acts as a counteracting cause, it moves in the opposite direction of δy. Since the objective is to explain δy, counteracting causes are not pursued further. The procedure continues to elucidate all contributing parsimonious variables until no additional contributing parsimonious causes remain.

Explanations based on a single equation from a business model M offer "one-level" explanations. However, to fully explain $\delta y = q$, it is advantageous to investigate the differences between the actual and reference values of its contributing causes. This enables further examination from one level to the next within the business model, aiming for a comprehensive explanation.

During this process, the distinct impacts of contributing and counteracting causes become evident. The focus is solely on the contributing causes for a comprehensive explanation, without delving into the counteracting causes. Generally, only the simplest and most straightforward causes are incorporated into the tree to ensure the explanation remains clear and robust.

3.1.3 Looking ahead and hidden causes

Cancelling-out or neutralisation effects occur when the positive results of the first quarter are cancelled out by the negative results of the second quarter. (Feelders and Daniels, 2001) developed a maximal explanation method that does not account for hidden causes, which can remain obscured due to the neutralisation by other variables. (Caron, 2013) developed a method to make hidden causes visible through substitution, enhancing the framework for generating explanation trees.

When explaining $\delta y = q$, the following equations from the business model M are used:

$$y = f(x) \in M_{p;(p+1)}.$$

$$x_i = g_i(z) \in M_{(p+1);(p+2)}.$$

where $x = (x_1, \ldots, x_i, \ldots, x_n)$ and $z = (z_1, \ldots, z_m)$ are n- and m-component vectors, respectively. These equations represent subsets of equations from the business model M, with $M_{p;p+1}$ expressing variables at level p using variables at level $p + 1$.

Suppose the explanation of $\delta y = q$ results in two sets of parsimonious causes, C_{bp} and C_{ap}, where $x_i \notin C_{bp}$ and $x_i \notin C_{ap}$, meaning x_i does not significantly influence y. To identify possible hidden causes, the children of x_i (elements of z in the second equation) are investigated by substituting the second equation into the first, resulting in:

$$y = h_i(x, z) \in M_{p;(p+2)}.$$

This process, called one-step look-ahead, generates explanations containing causes at level $p+2$ that might not be captured by maximal explanation. Finally below the formulas for contributing hidden causes as well as the hidden causes with counteracting and their influences on δy are provided.

Contributing Hidden Causes: Variable z_j is a contributing hidden cause if $z_j \in C_{bp}$ and $x_i \notin C_{bp}$, where z_j is a successor of x_i.
Counteracting Hidden Causes: Variable z_j is a counteracting hidden cause if $z_j \in C_{ap}$ and $x_i \notin C_{ap}$, where z_j is a successor of x_i.

The influence measure of z_j on y is given by:

$$\inf(z_j, y) = f(x^r_{-i}, g_i(z^r_{-j}, z^a_j)) - f(x^r_i, g_i(z^r)).$$

and the influence of x_i on y is given by:

$$\inf(x_i, y) = f(x^r_{-i}, x^a_i) - f(x^r) = f(x^r_{-i}, g_i(z^a)) - f(x^r_{-i}, g_i(z^r)).$$

These expressions show that the effect of z_j is neutralised by the effect of other variables in vector z.

In conclusion, the framework for generating explanation trees for business model M consists of maximal explanation and the extension for hidden causes. (Caron, 2013)introduced a one-step look-ahead algorithm to address the shortcomings of maximal explanation by investigating all non-parsimonious causes for potential neutralisation effects.

3.2 Informative Summarization

In this part, the second model called Informative Summarization, created by Sarawagi (2001) is discussed. She designed a method for explaining variations in aggregated functions within dimensional data, which can also be adapted for business dashboards that utilise a star database schema. Informative Summarization, originally intended for OLAP data, bears similarities to explanation formalism.

Informative Summarization employs the iDiff operator, which provides summarised explanations for changes in aggregated measures. Initially created for OLAP cubes, the iDiff operator examines the differences between two aggregated values, g_a and g_b. These values are derived by aggregating data across several dimensions. C_a and C_b denote the two sub-cubes created by expanding the aggregated dimensions of g_a and g_b. The cell values within these sub-cubes are represented by $v_a \subset C_a$ and $v_b \subset C_b$.

Sarawagi (2001)'s model organises cells with analogous changes to provide a concise summary of the disparities between C_a and C_b. This summarization process uses various methods, which depend on the selected grouping criteria, error functions, grouping formats, and objective functions.

The grouping criteria are used to assess the commonalities and similarities between the values v_a and v_b within C_a and C_b. These criteria often include absolute differences or ratios. Ratios are particularly effective for dimensions like geographical locations or product categories because they tend to have lower variance compared to absolute differences, making them a more reliable measure in these contexts.

The error function measures the discrepancy between the actual value v_b and the reference value derived from v_a using the summary statistic of its group. This function is defined as:

$$\mathrm{Err}(v_a, v_b, r) = (v_b - r \cdot v_a) \log\left(\frac{v_b}{r \cdot v_a}\right)$$

where r denotes the ratio of the cell groups containing v_a and v_b. This function emphasises significant changes by assigning more weight to both higher absolute differences and larger ratios, thereby highlighting the extent and proportion of change.

Handling missing values involves two scenarios, depending on whether v_a or v_b is missing. If v_a is missing, the ratio $\frac{v_b}{v_a}$ is maximised based on a user-defined factor F. Conversely, if v_b is missing, the ratio is minimised using the same factor.

The grouping format determines which cells are considered for grouping together. Dimensional hierarchies often create inherent divisions for these groupings, leading to clear and meaningful results.

There are two primary options for the objective function. The first involves specifying a fixed limit N on the number of rows in the final summary, with the aim of minimising total error. The second involves setting a fixed limit $e_{\max}$ on the error for each cell, with the goal of finding the smallest possible summary that meets this limit.

In conclusion, Informative Summarization, through the iDiff operator, offers a powerful tool for explaining differences in aggregated measures in OLAP systems. By grouping similar changes and employing comprehensive error functions, it effectively highlights significant changes that might otherwise be overlooked in large datasets. This approach, adaptable for business dashboards, ensures thorough and insightful analysis of data, aiding in better decision-making and understanding of underlying trends and factors.

3.3 Explanation by intervention

The third framework to be explored is the explanation by intervention algorithm, developed (Roy and Suciu, 2014)). Initially devised for SQL query applications within database systems, this framework centers around two principal concepts: intervention and causal paths.

Within this algorithm, an analyst's question is denoted as (Q, dir), where Q represents the numerical query and $dir \subset \{high, low\}$ indicates whether the user expects Q to be higher or lower than a reference value. Instead of a numerical query, Q can be considered an aggregated measure, particularly in business dashboards that utilise a star schema. The expression $Q = E(q1, \ldots, qm)$ shows how Q is constructed from various qi components, which are aggregate operators, and E represents an arithmetic formula consisting of mathematical operators.

A candidate explanation, ϕ, is characterised by a set of predicates applied to attributes:

$$\phi_j = [R_i, A \,\mathrm{op}\, c]$$

where R_i represents a relation, A represents an attribute within that relation, and op c$\{=, <, \leq$

, >, ≥} specifies the operator used.

The framework introduces two techniques for evaluating a candidate explanation: aggravation and intervention.

Aggravation: This technique assesses the value of Q for all subsets ϕ and ranks them based on their impact. The sign of a candidate explanation ϕ is determined by the direction of the symptom:

$$\mu_{\text{aggr}}(D, Q, dir)(\phi) = \begin{cases} -Q(D\phi) & \text{if } dir = \text{low} \\ Q(D\phi) & \text{if } dir = \text{high} \end{cases}$$

Thus, explanations with values even lower than $Q(D\phi)$ will rank higher. While straightforward, aggravation has its limitations as it may identify explanations that are minor but exhibit significant changes.

Intervention: This method involves removing certain measures from the database D. Intervention is represented as $\Delta = (\Delta_1, \ldots, \Delta_k)$, and $D' = D - \Delta$ is the database instance remaining after the intervention. The measure of impact is the extent to which the values in $Q(D')$ shift in the direction anticipated by the user:

$$\mu_{\text{interv}}(D, Q, dir)(\phi) = \begin{cases} Q(D - \Delta\phi) & \text{if } dir = \text{low} \\ -Q(D - \Delta\phi) & \text{if } dir = \text{high} \end{cases}$$

Interestingly, the sign for μ_{interv} is the inverse of that for μ_{aggr}. This difference arises because the goal is to alter the value of Q in the reverse direction of dir to address the problem.

In multidimensional problems, examining μ_{interv} or μ_{aggr} without discretion can lead to many redundant explanations. To tackle this issue, a filtering technique known as minimal append is used to eliminate redundant explanations. This process involves ranking the explanations in descending order according to their intervention or aggravation measures. Then, they are sequentially added to a new list, ensuring that redundancies are examined and removed during the process.

3.4 Explanatory Models Applications

The integration of explanatory analytics and business dashboard has been research extensively. (Askala, 2020), (Vink, 2020), (Spruijt, 2023) integrated the previous described models: explanation formalism, informative summarization, and explanation by intervention into a dashboard.

- Askala (2020) focused on implementing three models of explanatory analytics within Microsoft Power BI. The study concludes that these models significantly enhance dashboard functionalities, though there are areas needing improvement, particularly in visualisation and predictive modelling. User feedback highlighted a strong preference for visual representations, emphasising the necessity for more user-friendly interfaces. Among the models, the Explanation Formalism model is deemed the most suitable for software implementation due to its alignment with business dashboard principles and the effectiveness of its visualisations.

- Vink (2020) complements this by developing an artefact that automates diagnostic analytics, integrating it into business dashboards through dynamic R scripts. The primary conclusion is that diagnostic models are essential for generating explanations for data anomalies. Vink's artefact, which includes multi-level explanations and informative summarization, demonstrates how diagnostic analytics can be seamlessly integrated into dashboards using Python and R scripts. This approach allows for flexible implementation across various platforms, ensuring that business analysts can perform detailed diagnostic analysis interactively.

- Spruijt (2023) also concludes that integrating explanatory analytics with existing business intelligence systems can significantly improve diagnostics. The study also points out the current limitations of business dashboards in automated explanatory features and suggests leveraging existing infrastructures for improved diagnostics.

3.5 Generative AI

This part is about using text based Generative AI as an explanatory analytics model. First some basic information about text based generative AI is provided followed by how text based generative AI can be used as explanatory model.

Artificial Intelligence (AI) has as goal to create machine which can think and learn similar to the human intelligence. At its core lies a network of algorithms, data structures, and computational methodologies, orchestrating the intricate dance of information processing and decision-making that defines intelligent behaviour Holzinger et al. (2019).

A multitude of techniques converge to simulate human-like cognition, from classical rule-based systems to state-of-the-art deep learning architectures. These methodologies, collectively referred to as the internal framework of AI, form the backbone of intelligent systems, enabling them to perceive, reason, and act upon their environment in ways that mirror human intelligence.Central to this internal framework are neural networks, inspired by the structure and function of the human brain. Through layers of interconnected nodes, neural networks perform complex computations, extracting patterns and features from raw data. Deep learning, a subset of neural network-based approaches, has emerged as a dominant force in AI research, achieving remarkable success in tasks such as image recognition, natural language understanding, and game playing (Choi et al., 2020).

Generative Artificial Intelligence (AI) stands at the forefront of contemporary technological advancements, revolutionising various domains ranging from art and entertainment to healthcare and finance. By harnessing the power of machine learning algorithms, generative AI systems have the remarkable ability to create novel content, imitate human creativity, and even generate realistic images, videos, and text (Brynjolfsson et al., 2023).

Generative AI refers to a subset of artificial intelligence that focuses on creating new data instances from existing ones. Unlike traditional AI models that are designed for specific tasks, generative AI models operate in an unsupervised or semi-supervised manner, enabling them to generate content autonomously (Brynjolfsson et al., 2023).

A sub field of artificial intelligence is Natural Language Processing (NLP) is that focuses on enabling computers to understand, interpret, and generate human language. This field has witnessed significant progress with the advent of models like OpenAI's GPT (Generative Pre-trained Transformer) or Google's Gemini (Kang et al., 2020). These models leverage large-scale pre-training on vast text corpora to generate coherent and contextually relevant text across a wide range of tasks, including language translation, summarising, and dialogue generation. Through techniques like machine learning and deep learning, NLP models have achieved remarkable progress, enabling human-like interactions with machines and powering applications that revolutionise communication, information retrieval, and knowledge extraction (Cambria and White, 2014).

While all these features are promising, there are concerns regarding bias and fairness in generative AI models, as they often reflect the biases present in the training data. If not selected carefully, these biases can reinforce existing societal inequalities and harmful stereotype (Gichoya et al., 2023)

Furthermore, there is a concern regarding the phenomenon of "hallucination," wherein models produce content that diverges from the original data distribution and creates instances that do not accurately represent reality (McIntosh et al., 2023). Hallucination can occur due to various factors, including inherent biases in the training data, limitations in the model architecture, or insufficient diversity in the training samples (Zhang et al., 2023). This poses a significant challenge as it can lead to the generation of misleading or inaccurate content. Addressing the issue of hallucination requires a multifaceted approach that involves not only improving the diversity and representatives of training data but also enhancing the transparency and interpretability of generative AI models (Zhang et al., 2023).

For example, Teresa Kubacka, a data scientist, deliberately made up the term "cycloidal inverted electromagnon" and asked ChatGPT about this nonexistent concept. To her surprise, ChatGPT responded with a convincing explanation supported by credible-looking citations, prompting her to question whether she had accidentally typed in a real phenomenon (Bowman, 2022).

3.5.1 Generative AI As Explanatory Model

Generative AI can be used as explanatory model due to its ability to analyse large volumes of tables and texts to uncover hidden patterns that might not be immediately apparent through traditional analytical methods. By leveraging techniques such as neural networks, these models can process and learn from vast datasets, identifying intricate relationships and correlations (Wang et al., 2018).

Generative AI models can be trained on historical data to recognise patterns that signify normal behaviour within a system. For instance, in the financial sector, these models can analyse transaction data to identify typical spending behaviours of customers. By understanding what constitutes normal behaviour, generative AI can establish a baseline against which deviations can be measured (Wang et al., 2018). This ability is similar to the ability of the models described in the previous chapter.

(Thagard, 2024) examined the ability of ChatGPT to create and evaluate hypotheses, a process known as explanatory inference. He proposes benchmarks to assess ChatGPT's performance, finding it proficient in various fields such as science, medicine, law, and technology. ChatGPT demonstrates creativity through novel hypotheses and analogies and effectively evaluates them using criteria like explanatory breadth, simplicity, and predictive power.(Thagard, 2024) addresses conceptual challenges regarding explanation, understanding, causality, meaning, and creativity, arguing that ChatGPT's emergent skills from extensive training allow it to perform sophisticated explanatory inferences, thereby expanding the realm of intelligent reasoning to include AI systems.

3.5.2 Prompt Engineering

As discussed in the previous section hallucination is a well known problem when applying generative AI, prompt engineering is a way to deal with this problem. It is an increasingly important skill set when engaging with large language models (LLMs) like ChatGPT. In the context of artificial intelligence, particularly in systems like ChatGPT or other natural language processing (NLP) models, a prompt is essentially the input text that a user provides to initiate a response or reaction from the model. The prompt sets the stage for the interaction, guiding the AI on what to focus on and how to respond based on the language and context it is given.

The input of a prompt is the text that a user provides to the model. It signals what the user is looking for, whether it's an answer, a continuation of a text, a specific action, or a creative piece. It can include questions, commands, partial sentences to complete, scenarios to respond to, or any other form of textual input.The output of a prompt is the text generated by the model in response to the input prompt. It reflects the model's understanding and processing of the input. The output can vary widely depending on the prompt. It could be an answer to a question, completion of a started text, an elaboration on a topic, or any other relevant response. The process is depicted in figure 11. It is important to note that in this thesis, the Large Language Model (LLM) is not modified or adapted. Although it is possible to train an LLM, such as GPT-4, on a specific topic, this thesis utilises GPT-4 in its original form as a tool.

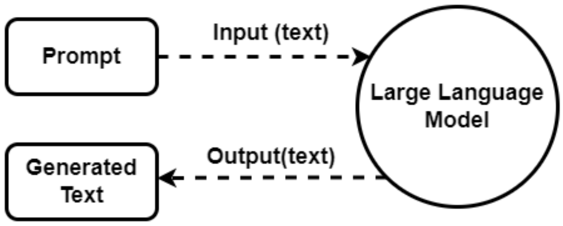


Figure 11: Input and output process of a NLP

Since the output heavily relies on the input of a prompt, it is crucial to understand the correct characteristics to achieve the desired outcome. According to the research by White et al. (2023), Zhou et al. (2022), (Wei et al., 2022) a good prompt possesses the following characteristics.

- Clarity and Specificity: The clearer and more specific a prompt, the more likely the NLP model generates a relevant and accurate response. Vague prompts can lead to ambiguous or unfocused responses.
- Length and Detail: Longer prompts that include more context can help the model understand the request better and provide more detailed and appropriate responses. However, overly long prompts can sometimes introduce unnecessary complexity or confusion.
- Directness: Prompts can be open-ended or directive. Open-ended prompts allow for more creativity and exploration in the model's responses, while directive prompts constrain the response to a more specific task or format.
- Bias and Framing: The way a prompt is phrased can influence the model's response, including potentially introducing bias. It's important to consider how the framing of a prompt might impact the response.
- Intent: The intent behind a prompt should be clear, especially for tasks that require a specific type of information or action from the model. This helps in aligning the model's response with user expectations.
- Adaptability: Effective prompts are adaptable to different contexts or slight variations without losing their effectiveness. This is especially important in dynamic environments where the context can change rapidly.

3.6 Comparison

In this section, the previously described models are discussed. Given that generative AI encompasses a broad range of technologies rather than a specific model, GPT-4 by OpenAI (2024) is the chosen model for comparison. The rationale behind choosing GPT-4 is detailed in section 4.7. The findings are summarised and shown in Table 1. This comparison builds upon the comparison done by (Askala, 2020). In this thesis the previous four explained models are compared on objective, algorithm, actual value, reference value, computational method, filtering method, input, output and constraints.

Objective

The first distinction is that Explanation Formalism, Informative Summarization and Explanation by Intervention have OLAP databases as objective making it extremely suitable for dashboard integration. However, GPT-4 was created to be used as a natural language processor, but as explained in chapter 3.5 it can also be used to analyse vast amounts of data.

Algorithm

The algorithm is different for each model. The Explanation Formalism by Feelders and Daniels (2001) and extended by Caron (2013) has a top-down method, it starts with the nearest dimensions and moves downwards based on contributing causes. In contrast the informative summarization model by Sarawagi (2001) uses the iDiff algorithm which processes tuples by hierarchy levels, updating partial solutions at each level from the bottom up. Each node in the hierarchy stores its best solution and passes it up to the next level, culminating in the optimal solution at the top node. The Explanation by Intervention from Roy and Suciu (2014) adopts a method,where all potential explanations are consolidated into a single comprehensive table. This allows for the simultaneous calculation of the impact of each possible cause. By evaluating all explanations together, the Explanation by Intervention model identifies the most significant factors by assessing the influence of each potential cause. GPT-4, diverges by using a transformer-based neural network, which relies on pre-trained data for natural language processing, rather than OLAP database-centric methods.

Actual Value

Each model interprets and utilises actual values in a distinct manner. For Explanation Formalism, the actual value is contingent on specific parameters or variables, denoted as r, within the data model to determine the true value at a given point. Informative Summarization treats the actual value as the true value at a given time, using the current, factual data point as the baseline for comparisons and summaries. Explanation by Intervention considers the actual value as the result of a database query, reflecting the data returned from querying the database at the present moment, which serves as the foundation for interventions and explanations. In contrast, GPT-4 generates the actual value based on the input context and its extensive training data. This approach allows GPT-4 to produce responses and explanations that are grounded in the user-provided context and the broad knowledge it has been pre-trained on.

Reference Value

Explanation Formalism bases its reference value on historical data and trends, using past data to establish a baseline or expected value for comparison with actual values. Informative Summarization follows a similar approach, relying on historical data and trends to summarise and compare current data points against established patterns. Explanation by Intervention derives its reference value from the result of a database query conducted at a different time than the actual value, allowing for a comparison of the current state of the data to a previous state to measure the impact of interventions. In contrast, GPT-4 uses a reference value generated from historical data, trends, patterns, and the extensive knowledge embedded in its pre-trained data.

Computational Method

Explanation Formalism calculates impact by considering the difference between actual and reference values and applying a specific function to these differences, determining how much the actual value deviates from the expected norm. Informative Summarization measures impact by evaluating both the magnitude and the ratio of differences between actual and reference values. This comprehensive approach assesses how significant the error is, based on the discrepancy between the current data point and the expected value. Explanation by Intervention uses a dual approach to measure impact, focusing on both aggravation and intervention. This method assesses how changes or interventions affect the data, considering whether these interventions increase or decrease the measured outcome, and the overall effect of removing or altering specific factors within the data set. For GPT-4, the measure of impact is determined by changes in model output, user engagement, and the relevance of responses. This involves evaluating how modifications in input prompts or context influence the responses generated by the model and assessing the significance and appropriateness of these responses based on user interactions.

Filtering Method

Explanation Formalism employs Fraction T, identifying and selecting data subsets based on a specified threshold to ensure that only the most relevant subsets are considered for further analysis. Informative Summarization uses a method where data is summarised and ordered by error, focusing on ranking data points based on their associated errors and prioritising those with the highest errors for detailed examination. Explanation by Intervention takes a different approach by ordering data based on the impact of interventions and creating minimal additions. This method priorities data according to the effects of specific interventions, making minimal necessary changes to understand cause-and-effect relationships. In contrast, GPT-4 utilises prompt engineering as its filtering method, which involves crafting specific input prompts to guide the model in generating the most relevant and contextually appropriate responses.

Input

Each model in this comparison requires different inputs to perform its analysis. Explanation Formalism, Informative Summarization, and Explanation by Intervention primarily rely on data or dashboards, as well as specific equations relevant to their methodologies.nOn the other hand, GPT-4 accepts a broader range of inputs, including not only data and equations but also user-generated prompts.

Output

The outputs produced by each model reflect their unique approaches and objectives. Explanation Formalism generates an explanatory tree, which visually represents the hierarchical relationships and contributing as well as the counteracting factors within the data. Informative Summarization produces a summarised table, highlighting key data points and their associated errors for easier interpretation. Explanation by Intervention results in an explanatory table that consolidates potential causes and their impacts, providing a comprehensive view of the data. In contrast, GPT-4 generates explanatory text, offering natural language responses that articulate the analysis and insights derived from the input data. This text-based output allows for a more intuitive and accessible understanding of the results, catering to users who may not be familiar with the intricacies of OLAP databases or advanced analytical models.

Constraints

Explanation Formalism has constraints aimed at preventing directional changes when multiple values are added, ensuring that the joint influence remains consistent. It introduces the conjunctiveness constraint to maintain direction and adds consistency to guarantee that actual and reference values meet the same functional requirements.Informative Summarization operates under additive constraints, summarising data by rolling up values to create higher-level hierarchies. This method relies on the additive nature of the data to simplify analysis.Explanation by Intervention places constraints on the dataset, requiring extensive foreign and primary key relationships and semi-join reduced databases. It combines additive and ratio constraints, comparing results on a relative scale to understand the strength of relationships through changes in other values. GPT-4 is constrained by data bias and context limitations. The model's output is influenced by the biases in its pre-trained data and the context provided by the user. Effective prompt engineering is essential to mitigate these limitations and ensure relevant and accurate responses.

Comparison Dimension	3.1: Explanation Formalism	3.2: Informative Summarization	3.3: Explanation by Intervention	3.5: GPT-4
Developers	Feelders and Daniels (2001) & Caron (2013)	Sarawagi (2001)	Roy and Suciu (2014)	OpenAI (2024)
Objective	OLAP Databases	OLAP Databases	OLAP Databases	Natural Language Processing
Algorithm	Top-Down	Bottom-Up	Big Bang	Transformer-based Neural Network
Actual Value	Depends on r	True value at a given time	Result of database query	Generated based on input context and training data
Reference Value	Historical, Trends, Statistical model	Historical, Trends	Result of database query where time ≠ actual value	Historical, Trends, Patterns and knowledge from pre-trained data
Computational Method	$\inf(X, y) = f(x_r - Ix, x_a Ix) - y$	$rErr(va, vb, r) = (vb - rva) \log \frac{vb}{rva}$	$\text{Aggravation}(D, Q, dir)(\phi) = \begin{cases} -Q(D\phi) & \text{if dir = low} \\ Q(D\phi) & \text{if dir = high} \end{cases}$ $\text{Intervention}(D, Q, dir)(\phi) = \begin{cases} Q(D - \Delta\phi) & \text{if dir = low} \\ -Q(D - \Delta\phi) & \text{if dir = high} \end{cases}$	Change in model output, relevance of responses
Filtering Method	Fraction T to filter parsimonious set, ranking	Summarise and order by Error	Order by Intervention, create minimal additions	Restriction in prompt
Input	Data or dashboard, equations	Data or dashboard	Data or dashboard	Data or dashboard, equations, prompt
Output	Explanatory tree	Summarised table	Explanatory table	Explanatory text
Constraints	Consistency & Conjunctiveness	Additive	Additive & Ratio	Data bias & context limitations

Table 1: Comparison of Explanation Formalism, Informative Summarization, Explanation by Intervention and GPT-4 (Askala, 2020)

3.7 Generative AI & Business Dashboards

As discussed, the integration of AI with business dashboards empowers organisations to make informed decisions, optimise operations, and enhance efficiency. In this section, current applications are analysed to assess the state of this integration. Presently, there are two primary applications of AI in conjunction with business dashboards: one involves using AI to design the layout of the dashboard, and the other involves leveraging AI to query the data. This analysis focuses only on the latter application, as it aligns with the thesis.

The surveyed applications were chosen based on popularity and availability. To check for popularity (Capterra, 2024) is used. Capterra is a company which assists consumers with selecting software for their needs with user reviews and research. Based from this Capterra's research the following ten companies were selected: Akkio (2024), Microsoft (2024a), ChatPowerBI (2024), Datamatics (2024), Turbonomic (2024), Involve (2024), Plotly (2024), Polymer (2024), Microsoft (2024b), Tableau (2024).

To assess the effectiveness of the surveyed applications, a test was created. This test includes five types of questions designed to evaluate the capabilities of each application. The first question is a value query / non computational query, a basic question to determine if the application is able to understand the table. The second is a comparison query, which is a bit more complex question to see if the surveyed application can compare the data in the table. The third query is a computation query / calculated measure which requires calculating something from the provided data. The fourth query is a factor identification query which requires identifying specific factors that contribute to a particular outcome, demonstrating a deeper understanding of the data. The fifth query is an Explanatory Query which requires understanding and interpretation of the data beyond simple calculation.

The first three questions are classified as descriptive queries which aims to provide straightforward, factual information about a specific subject or dataset. It typically involves summarising or describing characteristics, such as averages, totals, or distributions, without delving into reasons or causes behind the data. The last two questions are explanatory queries which seeks to understand the underlying reasons, causes, or mechanisms behind a particular phenomenon or observation. It involves interpreting data and providing insights into why certain patterns or trends occur.The explanatory questions are put in italics in table 2.

Below each question is the correct answer. The applications mentioned previously were asked the questions and examined for their answer. Since question 4 and 5 can be answered in multiple ways e.g. for question 4 the application could also have answered TA and PM which would also be correct. To avoid hallucination errors as described in 3.5.2 Prompt Engineering, before each question the sentence "base your answer solely on the provided data and formulas" was added.

The questions are based on the context provided by Feelders and Daniels (2001). There are two types of inputs: the data and the model/formulas, both of which are detailed in Appendix A.

Using these inputs and the question types defined in the penultimate paragraph, the following specific questions were formulated to compare the capabilities of the surveyed applications.

No.	Type of Question	Question	Answer
1	Value Query / Non Computational Query	What is the ROA of firm 1?	0.251
2	Comparison Query	Which firm has the highest ROA?	Firm 1
3	Computation Query / Calculated Measure	What is the average ROA?	0.137
4	*Factor Identification Query*	*Identify the factors that contribute to the ROA*	*TA (Total Assets) and PM (Profit Margin)*
5	*1-level Explanatory Query*	*Why is the ROA of firm 1 high compared to other firms?*	*Both TA & PM contribute to the relatively high ROA of firm 1*

Table 2: Questions to evaluate the integration of GAI in Business Dashboards.

The results of this test are presented in table 3. The first column contains the application surveyed, the second till sixth column represent Question 1 to Question 5 and the last cell is the source. A ticked cell means the application is able to answer the question and a blank cell means the application was not able to answer the question.

Application	Q1	Q2	Q3	Q4	Q5	Source
Akkio	✗					Akkio (2024)
Azure	✗	✗	✗			Microsoft (2024a)
ChatPowerBI						ChatPowerBI (2024)
Datamatics	✗	✗	✗			Datamatics (2024)
IBM Turbonomic	✗	✗	✗			Turbonomic (2024)
Involve	✗	✗	✗			Involve (2024)
Plotly	✗					Plotly (2024)
Polymer	✗	✗				Polymer (2024)
PowerBI	✗	✗	✗			PowerBI (2024)
Tableau	✗	✗	✗			Tableau (2024)

Table 3: Evaluation of the integration of GAI in Business Dashboards.

While most applications were able to perform adequately on the first three questions, a few applications under performed. Notably, ChatPowerBI returned an incorrect answer of 251 for question 1. Although Akkio and Plotly successfully answered question 1, they struggled with question 2, often providing hallucinated responses or failing to generate an answer from the provided information. Polymer managed to identify that firm 1 had the highest Return on Assets (ROA) but was unable to compute the average ROA.

All other applications had no difficulty with the first three questions, but none of the analysed applications could correctly answer neither question 4 or question 5. The responses were answers such as "Firm 1 has the highest ROA because it has the highest return on assets." or "No access to reasons or specified qualitative data" was returned.

The only two applications that came close to addressing question 4 were PowerBI and IBM Turbonomic. Both could generate an answer similar to the explanation tree - PowerBI's Decomposition Tree and IBM Turbonomic's Root Cause Analysis— but these tools required manual generation before they could answer the question. Also these applications are performing without the models in table 3. In conclusion, while generative AI can be useful in business dashboards for answering descriptive queries, it is not yet effective or used as a tool for answering explanatory queries.

4 Design & Requirements of the Explanatory Text Generator

This chapter outlines the design of the artefact, addressing the main research question: "How can the integration of artificial intelligence (AI) techniques enhance the explanatory analytics capabilities of business dashboards to improve business decision making?" In short, the artefact, henceforth referred to as the Explanatory Text Generator (ETG), aims to utilise artificial intelligence and explanatory analytics to enhance business decision making.

Based on the comparison of explanatory models in Chapter 3.6, the explanation algorithm was selected to generate explanations for the explanatory queries, which would be presented in text format via the OpenAI API. The explanation formalism was chosen because AI is employed in this thesis as a black box, necessitating that the inputs and outputs be verifiable. The study by Feelders and Daniels (2001) provides a limited dataset (only one table) that allows for manual calculations to evaluate the performance of the artefact. Consequently, only the explanation formalism by Feelders and Daniels (2001) is utilised, rather than the extension by Caron (2013).

Another rationale for selecting this algorithm is its visual output, in contrast to the tabular outputs of the informative summarization and explanation by intervention algorithms. Given that dashboards typically incorporate visual elements, this algorithm is the most advantageous from a business perspective. Furthermore, Askala (2020) noted in his research that the explanation formalism algorithm would be the most beneficial for a business dashboard.

Further in this chapter first, the requirements for the ETG are discussed, followed by the functional requirements which encompass more general needs. From there the input/output model is described in order to create the four UML (Unified Modelling Language) diagrams which are used to visualise system architecture, document design, facilitate communication, model software, and analyse requirements. By providing a common language for stakeholders, UML ensures clear communication and a shared understanding of system functionality and behaviour. This aids in the design phase by capturing detailed information about system architecture and specifications (Maciaszek, 2001). In this thesis, the use case diagram, class diagram, sequence diagram, and activity diagram are used.

The use case diagram captures and represents the functional requirements of the ETG, illustrating interactions between users (actors) and ETG functionalities (use cases). The class diagram models the static structure of the ETG, displaying its classes, attributes, methods, and the relationships among objects. The sequence diagram depicts the sequence of interactions and the flow of messages between objects over time, emphasising the dynamic behaviour of the system. The activity diagram represents the flow of activities or tasks within the ETG, highlighting control flow and the sequence of operations, which is particularly useful for modelling the logic of complex processes.

At the end of the chapter, after analysing the requirements, Input/Output diagram, UML diagrams, the choices of technology for this thesis are discussed.

4.1 Requirements

Creating an artefact that interacts with advanced AI systems demands a thorough comprehension of its needs. These requirements serve as the guiding principles for every stage of the artefact's creation, shaping its design and execution. First the specific requirements for this artefact are described after that follow the functional requirements for the artefact.

The specific requirements which refer to the specific abilities of the ETG. With the example of 1.2 in mind the following specific requirements are important:

- A text box where users can type questions.
- A classifier which determines whether it is a Descriptive Query or an Explanatory Query.
- If it is a Descriptive Query it requires the following:
  - Access to the DataBase (Excel file).
  - Access to the Business model equations.
  - Access to the OpenAI algorithm to analyse and generate answers.
- If it is an Explanatory Query it requires the following:
  - To be able to use the explanation formalism algorithm:
    - An input where users can set the parameters.
    - Which can read the business model and instantiate the variables.
    - Must be able to identify the parsimonious sets of causes.
    - Must be able to cut off counteracting causes and not analyse those further.
    - Must be able to transform the results into a dataframe.
  - Transform the dataframe mentioned in the previous bulletpoint to text by making use of GPT-4.
- For both cases it must have a display where the generated text is shown.

Functional requirements specifically describe what the system should do and outline the specific behaviour or functions of the system.

1. Natural Language Processing (NLP): The system must have the ability to generate and process natural language queries, supporting various question formats such as direct questions, queries, and commands. As discussed by (Jannach et al., 2021) this includes the capability to handle complex sentence structures, synonyms, and context-aware interpretations to ensure accurate comprehension of the user's intent. The NLP component should also be able to manage ambiguous queries by seeking clarification from the user when necessary, thereby improving the overall accuracy and relevance of the responses.

2. Data Query and Retrieval: The system must provide an interface to query and retrieve data from an Excel file. It should support operations such as filtering, sorting, and extracting specific data ranges or cells, ensuring efficient and accurate data access from the Excel file.

3. User-Friendly Design for Non-Technical Users: The system should be designed with a strong focus on usability for non-technical users. As stated by (Darejeh and Singh, 2013) this includes providing an intuitive navigation, and simplified features that do not require technical expertise. The interface should include clear visual aids to guide users through the process.

4. Answer Generation: The system must provide accurate and contextually relevant answers. Answers should be generated in a manner that is easily understandable, preferably in a standard format, to non-technical users, using clear and concise language. The system should offer explanations and justifications for the answers provided, ensuring users understand how conclusions were reached.

4.2 Input/Output Diagram

In this section, all the inputs and outputs for the ETG are detailed. Essentially, the ETG is a prompt builder that requires various types of input to construct the most appropriate prompt. The ETG can handle two types of queries: descriptive and explanatory. The primary difference between these queries lies in input 5, which is not needed for the Descriptive Query. Additionally, for the Descriptive Query, the Dashboard Data and the Business Model Equations are sent directly to the prompt. In contrast, for the Explanatory Query, these elements are first processed through the explanation formalism before being sent to the prompt. In figure 12 the input/output diagram is illustrated and below is a description of the diagram.

Input

1. Constraint Prompt: A fixed part of the prompt designed to maintain consistency in the answers and prevent 'hallucinations', as discussed in section 3.5.2. This ensures that all responses are aligned with the predefined criteria and helps maintain the integrity of the answers provided. The constraint prompt is specifically tailored to be different for descriptive and explanatory queries, ensuring appropriateness for various types of questions.

2. Query: The question that the user seeks to have answered. This query is the main driver for the interaction between the user and the system. It can either be a descriptive or Explanatory Query depending on the user's needs.

3. Dashboard Data: The information available on the dashboard that users can inquire about. This data is crucial as it provides the foundational knowledge needed for generating accurate and relevant responses. Users can leverage this data to gain insights and make informed decisions based on the information presented.

4. Business Model Equations: The equations that form the basis of the business model. These equations are essential for understanding the underlying mechanics of the business operations.

5. Explanatory Tables: Tables with the measure of influence and whether it is a contributing or counteracting cause generated by the explanation formalism, which include the following inputs:

   - Dashboard Data: as explained above.
   - Business Model Equations: as explained above.
   - Parameters: The parameters set by the user. These parameters allow users to customise the outputs based on their specific needs and scenarios. By adjusting these parameters, users can explore different outcomes and gain a deeper understanding.

Output

The output is the same, whether the user submits a Descriptive Query or a more detailed, Explanatory Query, the response provided comprehensively address the question, incorporating relevant data and explanations as needed. This approach ensures consistency and reliability in the answers, making it easier for users to understand and act upon the information provided.

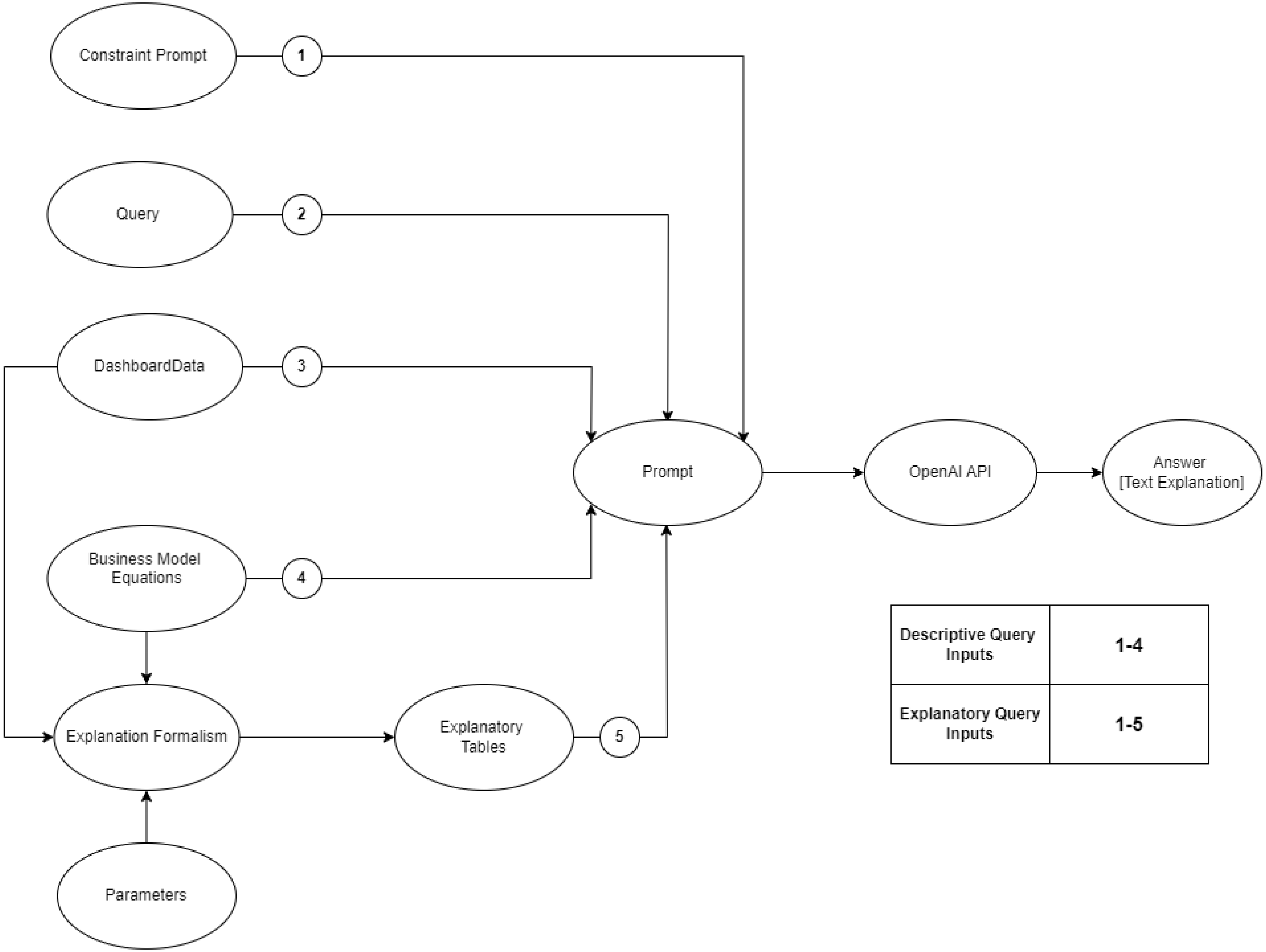


Figure 12: Input/Output Diagram of ETG (Explanatory Text Generator)

4.3 Use Case Diagram

A use case diagram is a visual representation used in software engineering and system design to illustrate the interactions between users (actors) and a system, showing the various ways in which a user might use the system. It helps in understanding the functional requirements of a system and clarifying how different components and processes are related. The main elements of a use case diagram include: Actors, Use Cases, Associations and System boundary. Actors represent the users or other systems that interact with the system being designed. Use cases depict the actions or services provided by the system to achieve a specific goal. Associations show the relationships between actors and use cases, indicating which actors participate in which use cases. The system boundary defines the scope of the system, indicating what is inside and outside of the system. Below are the actors and use cases described of this thesis Use Case Diagram.

ETG Actors

- *The Business Analyst*: the primary user who interacts with the ETG system to generate explanations.
- *The OpenAI API*: is an external system which is used to analyse data and transform prompts into textual explanations.
- *Dashboard Database*: is an external system which stores the data as well as the business model.

ETG Use Cases

- *Ask Query*: the business analyst initialises the use of the ETG by asking a query.
- *Classify Query*: the ETG classifies the question whether it is a Descriptive Query or an Explanatory Query.
- *Descriptive Query*: the ETG starts the Descriptive Query module. A Descriptive Query is a query which can also be answered by SQL.
  - *Build Prompt (Descriptive Query)*: the ETG starts to build the prompt for answering the Descriptive Query.
  - *Retrieve data (Descriptive Query)*: the data necessary for answering the Descriptive Query is retrieved.
- *Explanatory Query*: the ETG starts the Explanatory Query module. A Explanatory Query is a query which needs more information/context to generate an answer as opposed to the Descriptive Query.
  - *Set Parameters* the business analyst sets the parameters needed for the explanation formalism module.

- *Compute influences*: the influences of variables on the variable asked are computed following the procedure explained in section 3.1.1.
    - *Retrieve Data (Explanatory Data)*: the data necessary for answering the Explanatory Query is retrieved.
    - *Generate Filter Causes*: the causes which influence the outcome of a variable are filtered.
    - *Generate Explanatory Tables*: based on the causes the explanatory tables are generated.
    - *Build Prompt (Explanatory Query)*: the ETG starts to build the prompt for answering the Explanatory Query.

- *Generate Textual Explanation*: the ETG starts to generate the textual explanations by sending the prompts to the OpenAI API.

- *View Explanation*: the business analyst is able to see the current or previous explanation.

ETG Associations

- *Business Analyst* can interact with the following use cases: ask query, set parameters and view explanation.

- *OpenAI API* can interact with the following use cases: generate textual explanation and data analysis.

System Boundary

- *ETG(Explanatory Text Generator)*: encompasses the whole system.

- *Explanation Formalism*: encompasses the use cases compute influences, generate filter causes, generate explanatory tables which makes up the explanation formalism algorithm.

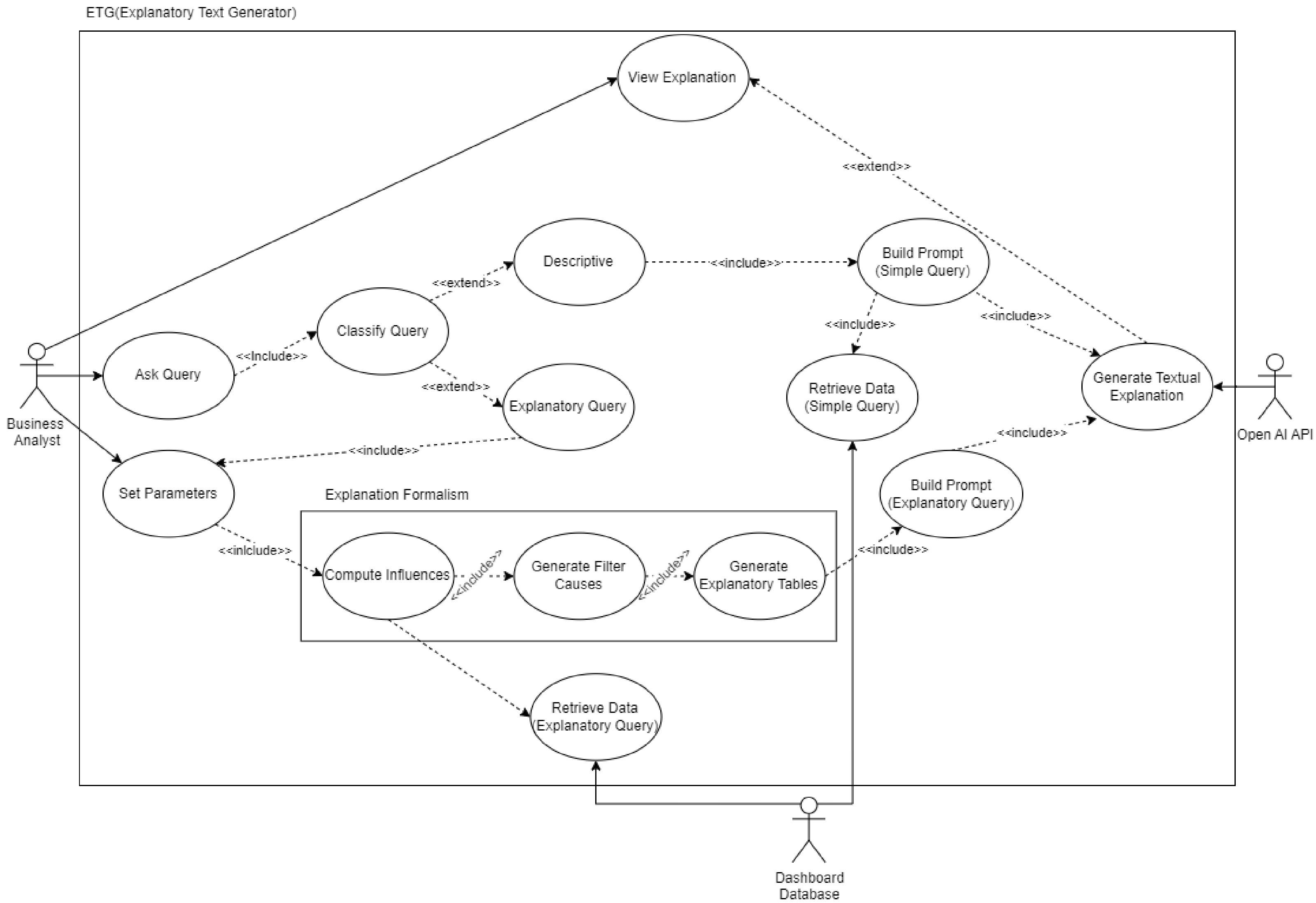


Figure 13: Use Case Diagram of ETG (Explanatory Text Generator)

4.4 Class Diagram

A class diagram models the static structure of a system by depicting its classes, attributes, methods, and the relationships among objects. It shows the blueprint of a system, illustrating how classes are interconnected and how they interact with each other. This helps in understanding the overall architecture, defining the properties and behaviours of different components, and establishing clear relationships and dependencies within the system. Classes represent the main building blocks, each defined with a name, attributes (properties or variables), and methods (functions or operations). Attributes define the characteristics or properties of a class, while methods specify the behaviours or functions that a class can perform. Relationships show how classes are connected and interact with each other. The ETG has five classes: ‘QueryClassifier‘, ‘GUI‘, ‘openAiApi‘, ‘DashboardDatabase‘, and ‘ExplanationFormalism‘ and sixteen methods.

1. *GUI*: managers the interaction between user and the rest of the system, it receives the queries, displays the answers and collects timestamps of both.

   - QueryView: This method retrieves a formatted view of the query asked by the user. It ensures the query is displayed in an easily readable format. The formatted view helps in better processing of the query.
   - Answer: This method shows the constructed answer based on the logged attributes. It ensures that the response is appropriately formatted and displayed. The user can see the reply generated by the system.
   - GetParameters: This method retrieves the parameters set by the user. It use a display with options for the user to select the appropriate parameters. The selected parameters are then used for further processing.

2. *Open AI API*: an important class which stores the API key in order to connect to the OpenAI API as well as building the prompts for sending to the API. It also contains the constraint prompt used to direct the answer.

   - GenerateAnswer: This method sends the Descriptive or Explanatory Query prompt to the OpenAI API. It generates a textual answer based on the provided prompt.
   - BuildPrompt: This method builds a prompt based on whether the query is descriptive or explanatory. For descriptive queries, it uses the query, data, business model, and the Descriptive Query constraint prompt to develop the prompt. For explanatory queries, it combines explanatory tables and the Explanatory Query constraint prompt to build the prompt.

3. *Dashboard Database*: manages database-related attributes and methods, storing information about the business model and the data in the dashboard database.

- getData: This method reads data from an Excel file. It processes the data and returns it for further use by the OpenAI API or the Explanation Formalism algorithm.
- getBusinessModel: This method retrieves the business model in the form of equations. These equations are used by the OpenAI API or the Explanation Formalism algorithm for further analysis. The business model helps in understanding the underlying logic of the queries.

4. *Explanation Formalism*: manages the computation of explanations by generating explanatory tables based on the set parameters by the user.
   - getData: This method imports the data to compute the measure of influence. It ensures that the data is available for the calculation process.
   - getBusinessModel: This method imports the business model to calculate the measure of influence. The business model provides the necessary framework for calculations.
   - ComputeInfluences: This method calculates the measure of influence based on the inputs of the data, business model, and parameters. It performs the necessary computations to determine the influence. The calculated influences are then used for further analysis.
   - GenerateFilterCauses: This method filters the causes based on the parameters and the calculated measure of influence. It ensures that only relevant causes are considered. The filtered causes are then used for generating explanations.
   - GenerateExplanatoryTable: This method generates explanatory tables from the filtered causes, which are tables in a predefined format. It makes the information comprehensible for GPT-4.
5. *QueryClassifier*: manages the classification of queries within the system. It can access the query and request parameters if necessary.
   - QueryView: This method accesses the query asked by the user. It ensures that the query is available for classification. The accessed query is then processed further.
   - ClassifyQuery: This method classifies the user input to determine if it is a Descriptive or Explanatory Query. It uses predefined criteria for classification. The classification helps in deciding the processing approach. Important to note in this thesis the classifier is not coded, there are two separate modules for answering either a Descriptive or Explanatory Query. In this thesis practical environment, the business analyst decides which module to use and is therefore the classifier.
   - RequestParameters: This method requests the necessary parameters from the user. It ensures that all required information is gathered. The requested parameters are then used for further processing.

Furthermore, the class diagram illustrates various associations that depict how the classes interact within the system. The 'Queryclassifier' class can have multiple instances interacting with a single 'GUI' instance, allowing numerous user messages to be classified. The 'GUI class has multiple instances interacting with a single 'openAiApi' instance, facilitating the processing of multiple user queries by the same API. The 'openAiApi' class has an optional relationship with the 'DashboardDatabase' class, where several API instances can interact with at most one database instance, ensuring efficient data retrieval and storage. Additionally, the 'openAiApi' class is associated with the 'ExplanationFormalism' class, where multiple API instances can utilise the formal explanation algorithms managed by a single 'ExplanationFormalism' instance. These associations ensure smooth data flow and cohesive operation within the system.

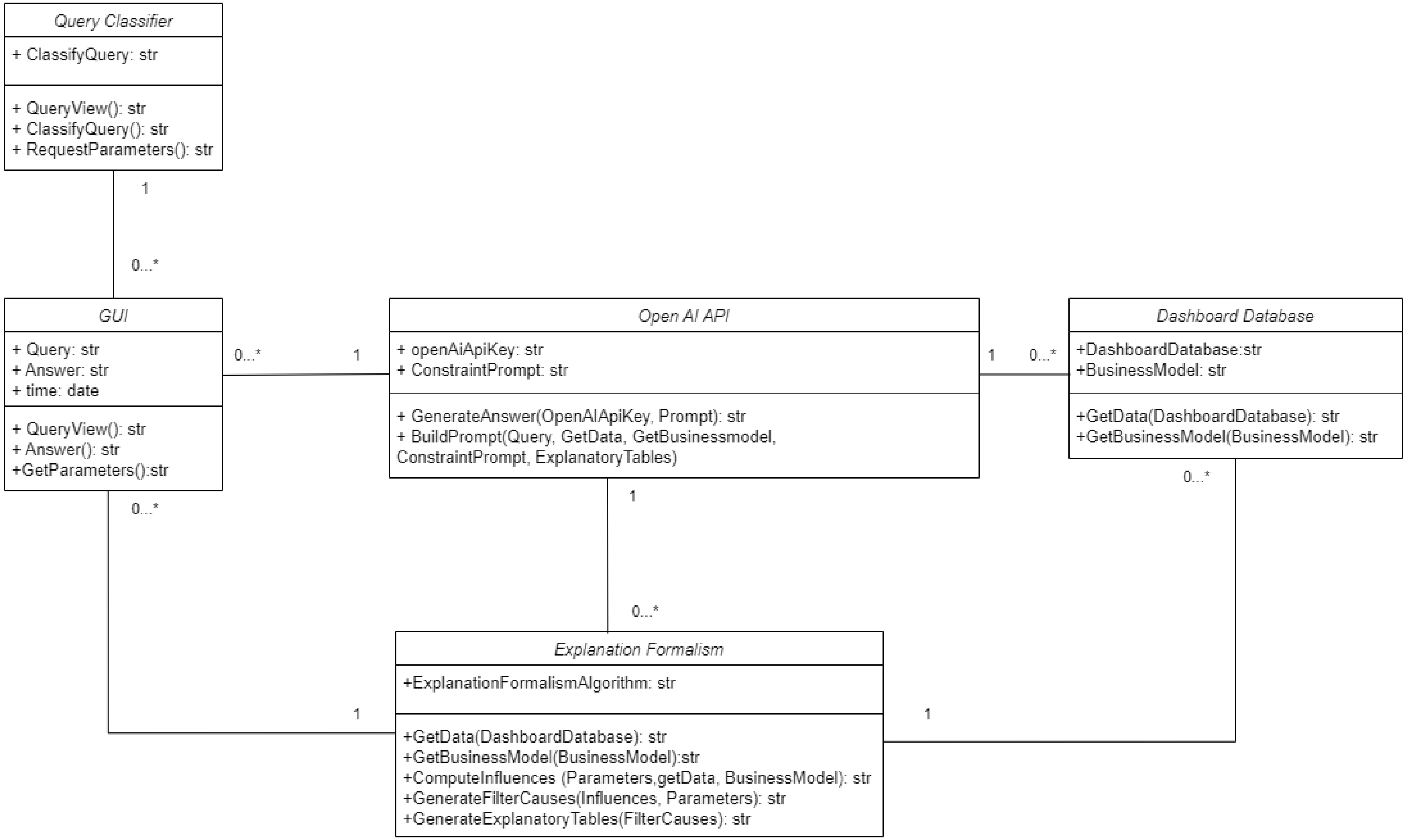


Figure 14: Class Diagram of ETG (Explanatory Text Generator

4.5 Sequence Diagram

A sequence diagram is a type of UML diagram that illustrates how objects interact with each other in a particular sequence of events. It focuses on the order of messages exchanged between objects to carry out a specific functionality or process within the system. The key components of a sequence diagram include objects (or lifelines), which represent instances of classes; messages, which are arrows indicating the communication between objects; activation or execution occurrences, which are the thin rectangles on lifelines showing the duration of an object's method execution. Messages are depicted as horizontal arrows between lifelines and represent the communication between objects. These messages can be synchronous, shown with solid arrows and filled arrowheads to indicate calls that wait for a response, or asynchronous, indicated by solid arrows with open arrowheads for calls that do not wait for a response. Return messages, depicted as dashed arrows, show the response from a called operation.

This thesis sequence diagram begins with the Business Analyst initiating the sequence by asking a Query, which is at that moment undefined. The Query is received by the 'GUI' component. The 'GUI' then forwards this request by sending a message to the 'QueryClassifier' to classify the query type.

Descriptive Query

The 'QueryClassifier' processes the request and if the query qualifies as Descriptive Query it sends the query to the 'OpenAIApi' to fetch an answer. Upon receiving this request, the 'OpenAIApi' retrieves and analyses the necessary data from the 'DashboardDatabase', generates an textual answer and returns it to the 'GUI'. The 'GUI' delivers the answer to the Business Analyst.

Explanatory Query

If the question is classified as an Explanatory Query additional parameters are needed, the 'GUI' requests these parameters from the Business Analyst. The Business Analyst sets the parameters and the 'GUI' sends them to the 'Explanation Formalism Algorithm'.

The 'ExplanationFormalismAlgorithm' then requests explanation data from the 'DashboardDatabase'. The 'ExplanationFormalismAlgorithm' generates explanatory tables and sends it to the 'OpenAIApi which generates a textual explanation and sends it to the 'GIU', which delivers it back to the 'Business Analyst'.

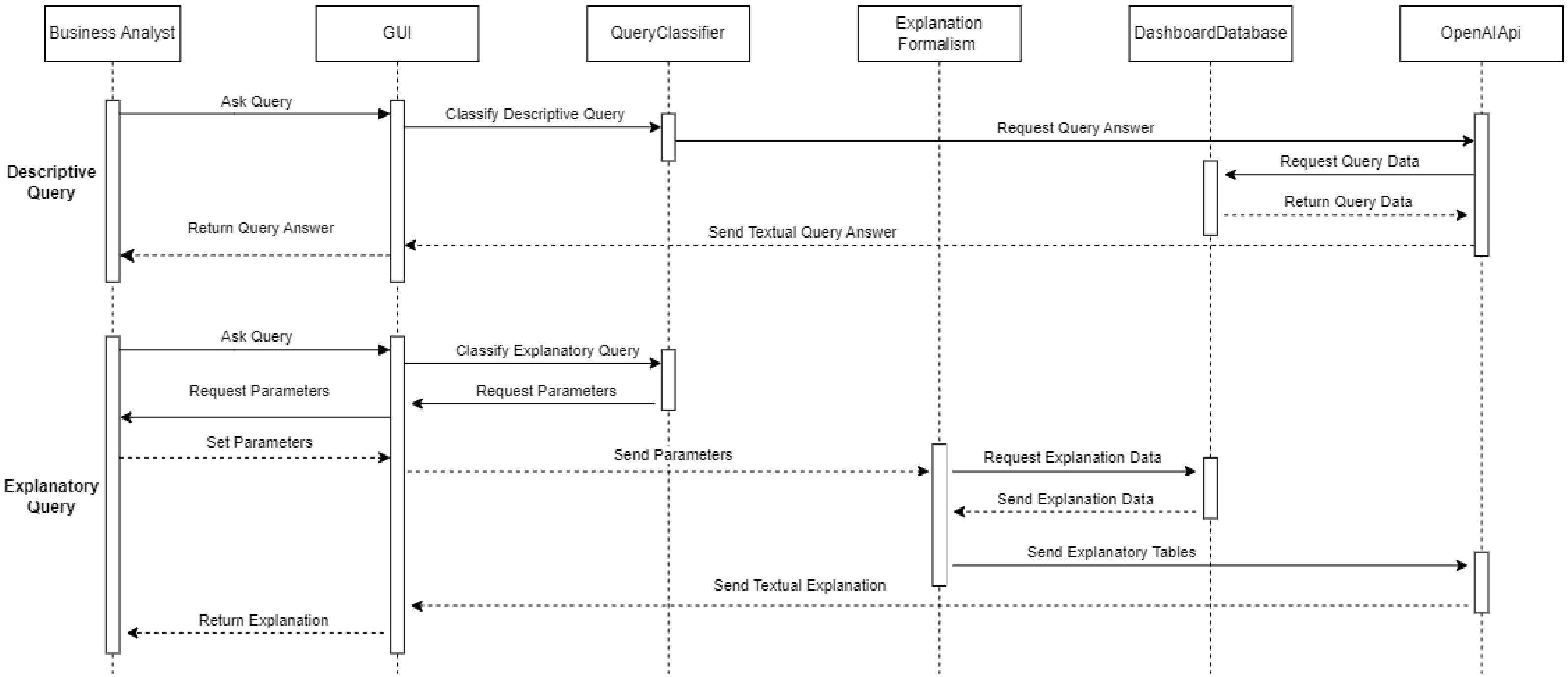


Figure 15: Sequence Diagram for Descriptive and Explanatory Query

4.6 Activity Diagram

An activity diagram is a type of UML diagram that represents the flow of activities or tasks in a system, highlighting the sequence and control flow of operations. It is used to model the logic of processes, workflows, and business rules. The main components of an activity diagram include activities, which are the tasks or actions taken; transitions, shown as arrows that indicate the flow from one activity to another; decision nodes, represented as diamonds that depict branching points with multiple possible path; initial nodes, shown as filled circles that indicate the starting point of the diagram; and final nodes, depicted as filled circles with a border, indicating the end point of the process.

Since most aspects have already been discussed in the previous chapters this text is kept short. The activity diagram illustrates the workflow starting with the Business Analyst asking a question via the GUI. The question is classified, and data(data as well as the business model) is fetched by the OpenAI API to generate an initial answer. If more details are needed, the system requests and the business analyst sets additional parameters. Further explanation data is processed to compute influences and causes, generating explanatory tables which are put into text by the OpenAI API. These detailed explanations are then returned to the Business Analyst through the GUI.

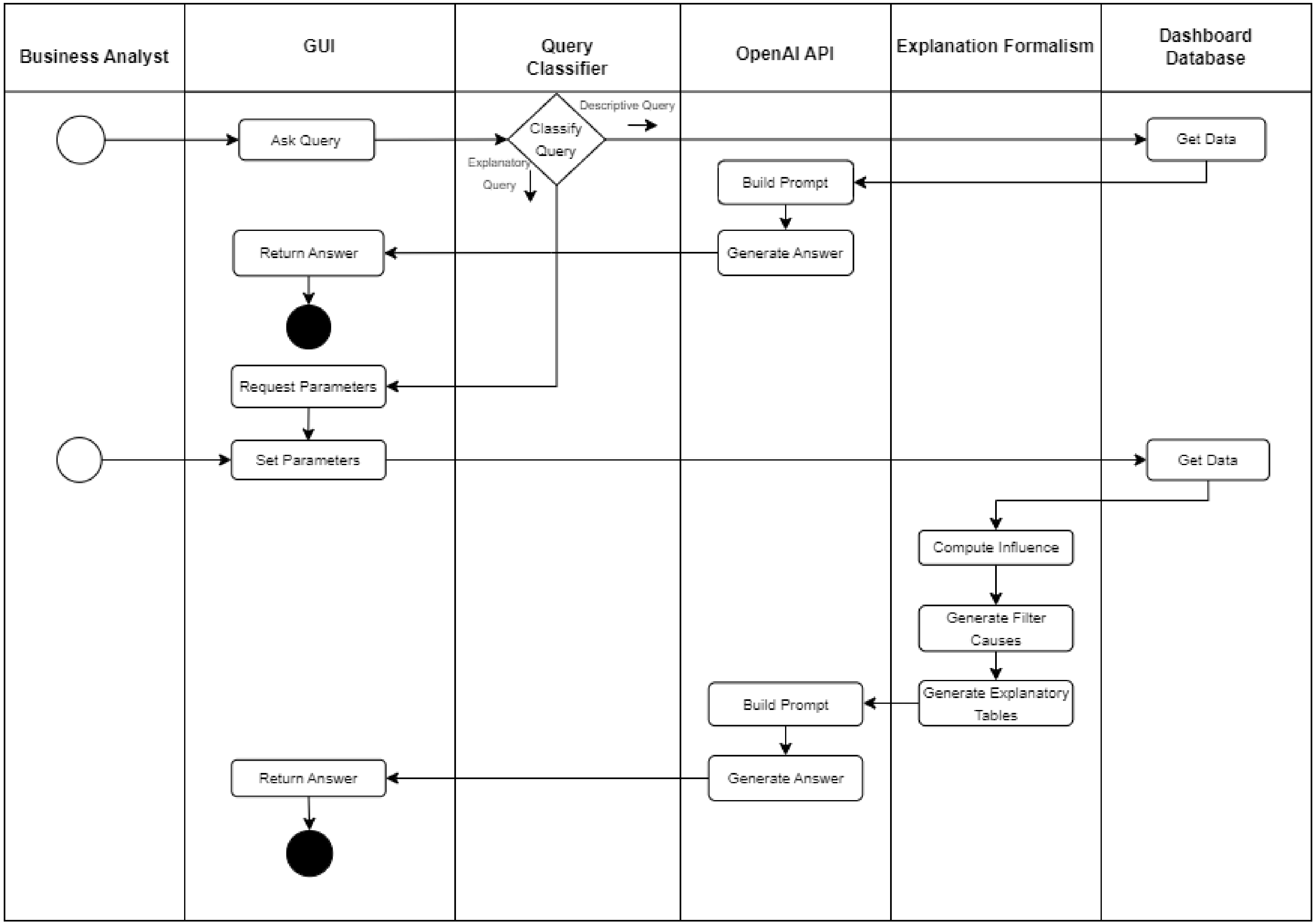


Figure 16: Activity Diagram for Descriptive and Explanatory Query

4.7 Choice of technology

This section explains the choice for Python as a programming language and GPT-4 as an AI tool for the thesis. Looking at things such as how versatile the language is, if many people use it, and if it matches with the project's requirements. For AI tools, considering how complex the tasks are, if there are ready-made models that can be used. By discussing about why these choices were made, this section helps understand the important thinking behind them, making it easier to see why these options were chosen.

Python

According to Github (2024) PYPL ranking, Python holds the top position. Java, JavaScript, C#, C/C++, and R follow Python in popularity. For the creation of the artefact, a data manipulation programming language is necessary, with Python or R being among the top 10 choices. Python stands out as a versatile language that excels not only as a steering language but also as a high-level language for scientific and engineering applications when enhanced with additional tools (Perez et al., 2010).

As argued by Perez (2007), Python's strength lies in its extensions, built upon useful general-purpose libraries. Python's straightforward object-oriented programming and clear syntax make it an accessible choice for data manipulation.

The combination of Python's clear syntax, its widespread use, and the Pandas library makes it an excellent tool for data processing, particularly in preliminary scientific research. Furthermore, this thesis employs the Visual Studio Code environment, a popular tool for interactive computing, data analysis, and visualisation across various scientific disciplines. Visual Studio Code offers a flexible and user-friendly platform for research and result sharing, making it an ideal choice for the purposes of this thesis.

AI

ChatGPT holds a prominent position among AI tools, it has a traffic share of 60%, reflecting its widespread adoption and effectiveness in diverse applications (statistica, 2024). Following the development trends, AI tools have been pivotal in advancing machine learning and artificial intelligence research (Kalyanathaya et al., 2019). For this thesis, a robust and versatile AI tool is necessary, with ChatGPT being among the top choices due to its advanced natural language processing capabilities.

Originally designed to generate human-like text based on the input it receives, ChatGPT has evolved significantly since its inception. The model's architecture, developed by OpenAI, leverages a variant of the transformer-based neural networks, which are particularly effective for understanding and generating language. This capability makes ChatGPT not only a powerful language model but also an excellent tool for engaging in dialogues, answering queries, and generating coherent, context-aware text in real-time.

The strength of ChatGPT lies in its adaptability and the vast dataset on which it has been trained. This enables the model to perform a wide range of language-based tasks, far beyond simple text generation. Its flexibility is evident in its deployment across various platforms and interfaces, where it enhances user interactions and improves accessibility to information through conversational AI.

The simplicity of integrating ChatGPT with other technologies also stands out. It can be interfaced with web applications, databases, and other tools to provide a seamless interaction layer. Moreover, the model's ability to continuously learn from interactions allows for improvements in accuracy and relevance of the responses over time, which is critical for maintaining engagement in user-centric applications, which is an important aspect of this thesis (Deng and Lin, 2022).

For data manipulation and analysis, ChatGPT's capabilities extend to interpreting and synthesising large volumes of text data efficiently. This makes it particularly valuable in fields requiring the extraction of insights from complex documents or user-generated content. Furthermore, the thesis utilises the Jupyter Notebook environment, which facilitates the integration of ChatGPT for interactive computing and natural language processing tasks. Jupyter Notebook offers a robust platform for executing live code, visualising results, and conducting iterative experiments, which enhances the research capabilities when combined with ChatGPT.

While a specialised GPT model such as the Business Intelligence GPT could be beneficial for this thesis because of its specialisation in the Business intelligence field. However, a specialised GPT model cannot be utilised for this thesis primarily because it lacks an API, which is crucial for seamless integration and practical application. Without an API, the specialised GPT model would require manual intervention for every interaction, significantly increasing the time and effort needed for data manipulation. This manual process is not only inefficient but also prone to human error, which can compromise the accuracy and reliability of the research findings. Moreover, the lack of an API restricts the ability to incorporate the model into existing workflows and systems, limiting its utility and scalability.

In conclusion, the selection of ChatGPT as the AI tool for this thesis is justified by its advanced linguistic capabilities, adaptability, and ease of integration with existing technologies. These features not only enable effective data manipulation and analysis but also contribute to the tool's utility in a broad spectrum of scientific and technical applications, thereby supporting the objectives of this research.

5 ETG Creation

This chapter delves into the development and functionality of the two types of queries within the ETG: Descriptive Query and Explanatory Query. Each query type is supported by two dedicated modules—one handling the graphical user interface (GUI) and the other focusing on the back-end logic for generating responses. These modules ensure a seamless interaction between users and the system, facilitating accurate and user-friendly query responses.

5.1 Descriptive Query

To address descriptive queries, two modules were developed: GUI Descriptive Query and Descriptive Query. As the names indicate, the first module handles the graphical user interface (GUI), while the second module is responsible for generating the answers. To facilitate understanding of the answer generation process, the Descriptive Query module is further divided into data preparation and prompt builder sections, where the key aspects of the code are thoroughly explained.

5.1.1 GUI Descriptive Query

For the GUI the Python package Tinker is used. Tinker is a Python package that provides a graphical user interface (GUI) toolkit, allowing developers to create user-friendly interfaces for their applications.Tinker simplifies the process of adding widgets such as buttons, labels, text boxes, and menus to applications, enabling interaction between the user and the program.

As this is a prototype, the emphasis was placed on developing a functional graphical user interface (GUI) rather than an aesthetically pleasing one. The Descriptive Query GUI is most aptly described as a chatbot, where users can input descriptive queries, and the interface subsequently displays the corresponding textual answers. In Appendix B the interface is depicted.

5.1.2 Data Preparation Descriptive Query

The Descriptive Query module starts by getting the data and the business model. When the ETG invokes the GetData and GetBusinessModel methods it starts the data preparation. This involves setting up the environment and loading the necessary data into the program. This begins with importing the Python libraries, which include *pandas* for data manipulation and the *OpenAI* library for natural language processing. The OpenAI API key is set to allow authenticated requests to the OpenAI service.

During *init*, several components are initialised. The *fixed_file_path* specifies the location of an Excel file that contains the data for analysis. The *formulas* variable holds the business model which are shown in Appendix A. The formulas were adapted to be used in Python e.g. in the original formula ROA was calculated as PM X TA or TA = 1 ÷ OA were changed to PM * TA and TA = 1 / OA.

An empty list named *conversation_history* is initialised to keep track of the interaction history. The *load_data* is called within the initialisation to read the data from the Excel file into a *pandas* Dataframe, which can be converted to be used by GPT-4.

The *load_data* tries to load the Excel file from the specified path into a DataFrame. If the data is successfully loaded, a confirmation message is printed. In case of any errors during this process, an exception is caught, and an error message is displayed, ensuring that the user is aware of any issues. The code for the data preparation part is in Appendix C.

5.1.3 Prompt Builder Descriptive Query

From there the Descriptive Query module starts building the prompt it involves constructing the interaction with the OpenAI API based on the loaded data and user inputs. The complete code for the Prompt Builder Descriptive Query is available in Appendix C. All components—the query, the constraint prompt, the data, and the business model—must be present to provide GPT-4 with the required context to answer the query accurately.

The order of constructing the prompt is crucial because it ensures that ChatGPT first understands the context before receiving the necessary data and equations. This sequence aligns with how instructions are typically processed—comprehending the task before analysing the details—thereby improving the accuracy of the response. As described in section 3.5.2 and stated by OpenAI (2024), below the components and order is explained.

send_chat: First the user message is loaded, this represents the query (number 2) in the input/output diagram in figure 12.

Constraint Prompt: Then OpenAI receives a constraint prompt that informs it about the data reception. This prompt instructs OpenAI to use only the provided data and equations, emphasising the importance of keeping the response concise, discussed in section 3.5.2. This part of the prompt represents the constraint prompt (number 1) in the input/output diagram in figure 12.

Data and Formulas: Subsequently, *data_prompt* and *self.formulas* are loaded, which include the necessary data and the business model. This represents number 3 and 4 in the input/output diagram in figure 12.

Prompt Builder Descriptive Query

```
def send_chat(self, user_message):
        if user_message:
            try:
                self.conversation_history.append({"role": "user", "content":
                ↪ user_message})
```

```
            if not any("Excel Data:" in message["content"] for message in self.conversation_history):
                data_prompt = self.df.to_string(index=False)
                analysis_prompt = f"You are given the following data. Use only the provided data and the formulas, it is important to give only the information that is asked to keep the answer as short as possible. :\n{data_prompt}\n\n{self.formulas}"
```

5.1.4 Output Descriptive Query

The ETG's output for a Descriptive Query is a concise textual explanation, as the nature of descriptive queries does not necessitate lengthy responses. This brief explanation aims to clarify the asked phenomenon in a straightforward and easily understandable manner. The brevity ensures that users can quickly grasp the essential information without sifting through extensive details. In Appendix C is an example of such an output to illustrate how the system generates these succinct explanations.

5.2 Explanatory Query

To address the Explanatory Query, two modules are developed: GUI Explanatory Query and Explanatory Query. Following the same principle as the Descriptive Query, the first module implements the graphical user interface, while the second module generates the answers. The code is adapted to work with the dataset in Appendix A. To facilitate understanding, the Explanatory Query module is divided into three sections: data preparation, explanation formalism, and prompt builder. In the Output section the HTML aspect is quickly explained. The complete code for the explanation formalism is provided in Appendix D, with the key parts explained in the subsequent sections.

5.2.1 GUI Explanatory Query

There is no visual GUI for the Explanatory Query. Instead, it consists of a list of selectable variables. Checks have been implemented to prevent errors such as selecting the wrong firm, dimension, variable, unknown direction, or incorrect threshold values. Examples of these errors include: selecting firm 11 and dimension 5 when DMC is requested, making a typo in the selected variable, choosing a variable without a formula like SPP, typing errors in specifying high or low, and selecting a threshold value less than 0 or greater than 1.

GUI Explanatory Query

```
# Input variables
selected_firm = 'Firm 9'
dimension = 5
selected_variable = 'ROA'
direction = 'high'
threshold = 0.85
output_file = 'final_report.txt'
```

5.2.2 Data Preparation Explanatory Query

The data is loaded from the Excel file, which contains the data of Appendix A. The code retrieves the values for variables like ROA for a specific firm from the data. This is done by setting the 'Variables' column as the index and then accessing the firm's values.

Data Preparation Explanatory Query

```
# Set the 'Variables' column as the index to retrieve specific values for firms
firm_values = data.set_index('Variables')[selected_firm].to_dict()
industry_values = data.set_index('Variables')['Industry Average'].to_dict()
```

5.2.3 Explanation Formalism

The Explanation Formalism part of the code starts by computing the influence of various variables on the selected variable for a firm, considering the direction (high/low) and comparing it to industry averages. Generally the direction is multiplied by -1 when the direction is low except for a few variables e.g. TA is calculated by dividing 1 / OA, this means that when the OA is high the TA is low and therefore the influence sign is -1 when the direction is high except when OA or it's variable components are calculated. The variable is labelled as contributing if the influence is more than 0 and counteracting if the influence is less than 0.

Calculate Measure of Influence

```
def calculate_firm_metrics(data, firm_name, selected_variable, direction, threshold):
    firm_values = data.set_index('Variables')[firm_name].to_dict()
    industry_values = data.set_index('Variables')['Industry Average'].to_dict()
    influence_sign = -1 if direction == 'low' else 1

    influences = {}
    if selected_variable == 'ROA':
        influences['Influence_PM_ROA'] = influence_sign * round((firm_values['PM'] *
        ↪ industry_values['TA']) - industry_values['ROA'], 4)
        influences['Influence_TA_ROA'] = influence_sign * round((firm_values['TA'] *
        ↪ industry_values['PM']) - industry_values['ROA'], 4)
    # Rest of the code for calculated influences is in Appendix D.

    def add_influence_result(variable, industry_value, firm_value, influence):
```

```
        return {
            'Firm': firm_name,
            'Variable': variable,
            'Industry Average': industry_value,
            'Firm Value': firm_value,
            'Measure of Influence': round(influence, 4),
            'Contributing/Counteracting': 'Contributing' if influence > 0 else 'Counteracting'
        }

    results = [add_influence_result(var_name, industry_values[var_name], firm_values[var_name], influences[influence])
               for influence in influences]

    df = pd.DataFrame(results)
    return df
```

Then the parsimonious sets are identified with the `calculate_parsimonious_set` function when the smallest set of influences that collectively meet or exceed a threshold (`threshold`). It starts with individual influences and then checks combinations if needed.

Create Parsimonious Sets

```
def calculate_parsimonious_set(df, threshold, effect_type):
    total_influence = df[df['Contributing/Counteracting'] == effect_type]['Measure of Influence'].sum()
    df['Percentage of Influence'] = df.apply(
        lambda row: round((row['Measure of Influence'] / total_influence), 4) if row['Contributing/Counteracting'] == effect_type else '',
        axis=1
    )
    vars_effect = df[df['Contributing/Counteracting'] == effect_type]
    individual_effect = vars_effect[vars_effect['Percentage of Influence'].astype(float) >= threshold]

    if not individual_effect.empty:
        return individual_effect
    else:
        for r in range(2, len(vars_effect) + 1):
            for combo in itertools.combinations(vars_effect.index, r):
                if vars_effect.loc[list(combo), 'Percentage of Influence'].astype(float).sum() >= threshold:
                    return vars_effect.loc[list(combo)]
    return pd.DataFrame()
```

When the parsimonious sets have been created the counteracting variables are removed in order to meet the criteria set in section 3.1.2 i.e. only contributing variables are in the generated report. Furthermore the `get_missing_variables` function ensures that variables which are not part of either the contributing or counteracting parsimonious set are not appearing in the generated report.

```python
def filter_variables(all_counteracting_variables, fixed_list):
    return [var for var in all_counteracting_variables if var in fixed_list]

def get_missing_variables(filtered_counteracting_variables, filtered_dimensions):
    missing_variables = set()
    for key in filtered_counteracting_variables:
        if key in filtered_dimensions:
            missing_variables.update(filtered_dimensions[key])
    return list(missing_variables)
```

The `generate_report` function filters variables and creates explanatory tables, which are then used by GPT-4. Each iteration is titled based on the user's query. figure 17 shows a table for a one level explanation. Multiple tables are used for multi-level explanations, as seen in figure 18. Adjusting the threshold changes the table, removing variables which are not part of the contributing or counteracting parsimonious set.

```
The factors on why the ROA of Firm 3 is high

Variable Explained: ROA

  Firm Variable  Industry Average  Firm Value Measure of Influence Contributing/Counteracting
Firm 3      ROA             0.137       0.189
Firm 3       PM             0.120       0.151               0.0351               Contributing
Firm 3       TA             1.140       1.250                0.013               Contributing
```

Figure 17: Python Explanatory Table for firm 3, Variable = ROA, T=0.85, Direction = high, Dimension = 1

The factors on why the ROA of Firm 3 is high

Variable Explained: ROA

Firm	Variable	Industry Average	Firm Value	Measure of Influence	Contributing/Counteracting
Firm 3	ROA	0.137	0.189		
Firm 3	PM	0.120	0.151	0.0351	Contributing
Firm 3	TA	1.140	1.250	0.013	Contributing

Variable Explained: PM

Firm	Variable	Industry Average	Firm Value	Measure of Influence	Contributing/Counteracting
Firm 3	PM	0.120	0.151		
Firm 3	AC	0.079	0.082	-0.003	Counteracting
Firm 3	PC	0.722	0.711	0.011	Contributing
Firm 3	DMC	0.072	0.047	0.025	Contributing
Firm 3	RDC	0.007	0.009	-0.002	Counteracting

Variable Explained: TA

Firm	Variable	Industry Average	Firm Value	Measure of Influence	Contributing/Counteracting
Firm 3	TA	1.140	1.25		
Firm 3	OA	0.877	0.80	0.11	Contributing

Figure 18: Python Explanatory Table for firm 3, Variable = ROA, T=0.85, Direction = high, Dimension = 2

5.2.4 Prompt Builder Explanatory Query

Compared to the Descriptive Query prompt builder, the Explanatory Query is slightly more straightforward. This simplicity arises from the lack of conversation history. The full code for the prompt builder explanatory query is in Appendix D, below the most important part of the code is displayed.

All components—the query, the constraint prompt, the data, the business model—must be present to provide GPT-4 with the required context to answer the query accurately. The order of constructing the prompt is crucial because it ensures that ChatGPT first receives the context before receiving the explanatory tables. This sequence aligns with how instructions are typically processed—comprehending the task before analysing the details—thereby improving the accuracy of the response. As described in section 3.5.2 and stated by OpenAI (2024), below the components and order is explained.

Constraint Prompt: First, OpenAI is taught its role as "a helpful assistant" and provided with relevant abbreviations e.g. ROA = Return on Assets, every abbreviations is in Appendix D. Then it is given the prompt that it should not mention the table or measure of influence to ensure a good narrative. It is instructed that the first variable is the selected variable in order to ensure that the contributing and counteracting causes are explained about the correct variable. This is all the information that is required ensures that no additional information is used for the textual explanation. The constraint prompt is number 1 in the input/output diagram in figure 12.

Report: the query, dashboard data, Business Model Equations and explanatory tables, respectively number 2-5 in the input/output diagram in figure 12 are all in the report. The title represent the query of the user. The data and business model have been used to determine the explanatory tables.

Part of Prompt Builder Explanatory Query

```
def explain_report(report):
    response = openai.ChatCompletion.create(
        model="gpt-4",
        messages=[
            {"role": "system", "content": "You are a helpful assistant. The following abbreviations will be used:\n" + abbreviations},
            {"role": "user", "content": f'''Explain, without mentioning the table or measure of influence, why the variable is high or low by indicating the contributing and counteracting factors from the table.
            Note that the first variable is the selected variable and that all other tables when referring to contributing or counteracting refer to the influence on the selected variable.
            This is all the information that is required.
            This is the table:\n\n{report}\n\nExplanation:'''}
        ],
```

5.2.5 Output Explanatory Query

The ETG's output for an Explanatory Query is a detailed textual explanation presented as a business report. A simple Python script was written to convert the text generated by the OpenAIApi and convert it to HTML.

The requested parameters are displayed prominently as a bold title. The explained variables are first sorted per dimension and then per equation. The explained variable is put as header above the paragraph. The explained variables are in italics with a red background, the words contributing and counteracting are blue and yellow respectively. The narrative format of the text ensures that it does not refer to tables or measures of influence, maintaining a seamless flow. An example of this output, using the table in figure 18 as input, is provided in 19. This text should contain the information of the explanatory tree in figure 20

The factors on why the ROA of Firm 3 is high

Return on Assets (ROA)

The high Return On Assets (ROA) of Firm 3 can be attributed primarily to two key factors: *Profit Margin (PM)* and *Turnover Assets (TA)*. The firm's PM is notably higher than the industry average, indicating that the company is able to generate a significant level of profit for each dollar of revenue it earns. This higher profitability can increase the company's overall ROA, as it suggests the firm is utilizing its assets more effectively to generate profit. Similarly, an above-average TA can also contribute to a higher ROA. An increased TA means the firm is using its assets more efficiently to generate sales. In summary, Firm 3's high ROA is mainly due to its superior ability to generate profit and efficient use of its assets compared to other firms in the industry. This performance can be seen as a result of a well-implemented business strategy and effective resource allocation.

Profit Margin (PM)

The Profit Margin (PM) of Firm 3 is higher than the industry average, driven primarily by two *contributing* factors. Firstly, the firm has a lower *Distribution and Marketing Costs (DMC)* compared to the industry average, which means Firm 3 spends less on this aspect, raising their net income and thus, *contributing* positively to a higher PM. Secondly, Firm 3's *Production Cost (PC)* is also lower than the industry average, again saving Firm 3 funds and boosting profits. However, there are also *counteracting* components impacting the PM. Firm 3 has slightly higher *Administrative Costs (AC)* compared to the industry's norm, which eats into their gross profit, reducing their PM. It's also worth noting that Firm 3's *Research and Development Costs (RDC)* are slightly higher than the industry average, meaning they invest more in their business, but in the short term, this higher expenditure acts as a *counteracting* factor, lowering their PM. The interplay of these *contributing* and *counteracting* elements results in a high overall PM for Firm 3 compared to the industry average.

Turnover Assets (TA)

The Turnover Assets (TA) of Firm 3 is high when compared to the industry average. The primary *contributing* factor to this higher value of TA seems to be the *Operating Assets (OA)* of the company. The OA is lower than the industry average and therefore it is *contributing* to the TA. A lower value of OA results in higher efficiency in asset utilization, thereby increasing TA. This means that the firm is relatively good at utilizing its assets to generate revenues, which leads to a higher TA.

Figure 19: Explanatory Text for firm 3, Variable = ROA, T=0.85, Direction = high, Dimension = 2

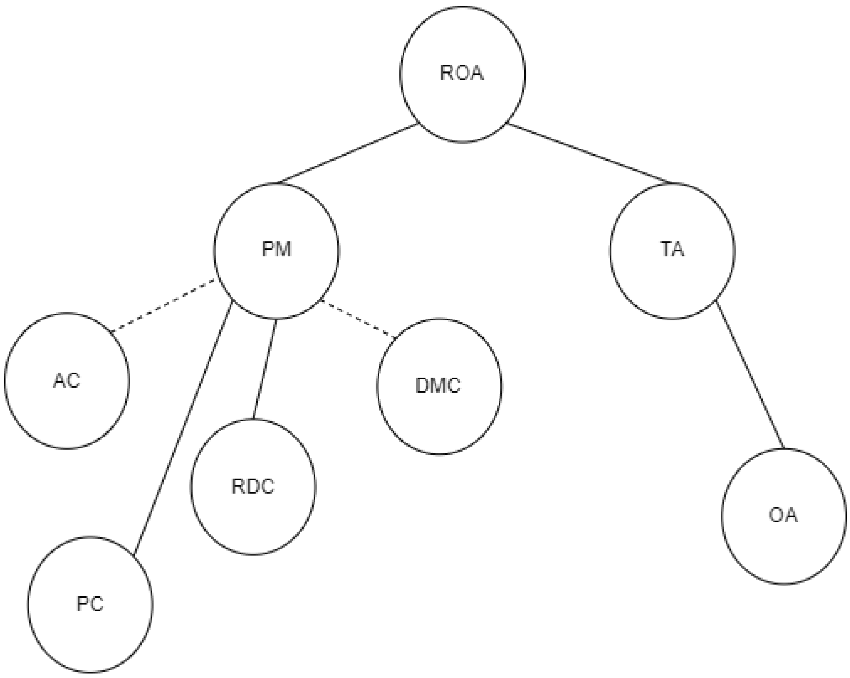


Figure 20: Explanatory Tree for firm 3, Variable = ROA, T=0.85, Direction = high, Dimension = 2

6 Evaluation

The evaluation includes a cognitive walkthrough of the Explanation Text Generator (ETG). A cognitive walkthrough is a usability evaluation method where evaluators simulate a user's experience to identify potential usability issues in an interface design. This process involves creating task scenarios and analysing the steps users take to complete these tasks. Evaluators assess whether users can achieve their desired goals, notice available actions, associate actions with intended outcomes, and recognise progress (Mahatody et al., 2010). The observations are recorded and for the descriptive queries compared to the test created in section 3.7. For the explanatory queries, first the results of the inputs are compared with manual calculations. Then the output of both methods is compared (explanatory tree and textual explanation) and lastly an operational comparison is made for each output's advantages and disadvantages.

6.1 Descriptive Query

For the Descriptive Query the test scenarios are the value query, comparison query and computation query of section 3.7 which are evaluated.
The value query, comparison query and computation query where entered into the GUIDescriptiveQuery module, the results are in Appendix B.

1. What is the ROA of firm 1?
   - Pre-defined answer: 0.251.
   - ETG answer: The ROA of firm is 0.251.
   - The difference with the pre-defined answer: The answers are the same. However, the ETG provides additional context.
2. Which firm has the highest ROA?
   - Pre-defined answer: 0.251
   - ETG Answer: Firm 1 has the highest ROA with a value of 0.251.
   - The difference with the pre-defined answer: The answers are the same; however, the ETG provides additional context by specifying the firm with the highest ROA.
3. What is the average ROA?
   - Pre-defined answer: 0.137.
   - ETG Answer: The industry average ROA is 0.137.
   - The difference with the pre-defined answer: The answers are the same; however, the ETG provides additional context by explicitly stating that the value represents the industry average ROA.

The conclusion is that the Descriptive Query ETG effectively answers descriptive queries and can even provide additional context. As demonstrated in Appendix B, it can also address more complex queries, such as identifying the factors that contribute to the ROA, which falls under explanatory queries.

6.2 Explanatory Query

For the Explanatory Query, it is important that the inputs of the ETG are consistent with those of the Explanatory Tree to enable accurate comparison of the outputs. This is crucial due to the use of AI as a black box in this thesis. Therefore, it is necessary to carefully verify that the inputs are correctly determined before comparing the outputs. Initially, the inputs are calculated to ensure the code generates the same results. These inputs are checked through one-level analysis with various variables, multi-level analysis, different thresholds, and different directions.

Then the outputs are explained i.e. the explanatory tree is compared against the textual response to see if it contains the correct information. Lastly an operational comparison is made between the outputs and their advantages and disadvantages are discussed.

6.2.1 Explanatory Query Input

Starting with the one level analysis. For every variable a random firm was selected and the input's were calculated and compared against the results of the code. In appendix F is an example for firm 7's FA which was also chosen randomly. The final results of the comparison is in table 4 where a check mark means the code identified the same results as the manual analysis a cross means that the code was not able to do it.

Variables	Firm1	Firm2	Firm3	Firm4	Firm5	Firm6	Firm7	Firm8	Firm9
ROA		✓							
PM								✓	
TA	✓								✓
OA							✓		
PC									✓
DMC				✗					
MC			✓						
CA					✓				
FA									✓
WIP							✓		
FGS						✓			
DEB				✓					

Table 4: Explanatory tables of the code examined for one level analysis.

After completing the one-level analysis, the performance of the multi-level analysis was evaluated. The ROA with a dimension of 5 and a threshold of 0.85 was calculated for each firm. The direction was appropriately set: if the firm's value exceeded the industry average, the direction was marked as high, and if it was below, it was marked as low. Appendix F provides an example for firm 2, which was selected at random, of how this input was verified. The results of this test is in table 5.

Variables	Firm1	Firm2	Firm3	Firm4	Firm5	Firm6	Firm7	Firm8	Firm9
ROA	✓	✓	✓	✓	✓	✓	✓	✓	✓

Table 5: Explanatory tables of the code examined for multi level analysis.

Based on the one-level and multi-level tests, it can be concluded that the code produces the same results as manual calculations which proves its ability to generate the intricate relationships of the model. It accurately measures influences, classifies contributing and counteracting factors, handles variables that are zero, and creates the same parsimonious set.

However, a few inaccuracies were noted. The measure of influence for qualitative functions is not accurate. Since these relationships are qualitative, this measure holds no value, especially for the variables MU and SPP, where numbers represent nominal scales (e.g., MU = 1 indicates the firm predominantly used ferrous metals, while MU = 2 indicates nonferrous metals). This measure of influence is still needed to determine whether a variable is contributing or counteracting. This causes no issues in the ETG output because, as explained in Section 5.2.4, the measure of influence is not included in the ETG output.

Additionally, the ETG's explanatory tables cannot handle cases where a firm's value equals the industry average or reference value. In multi-level analysis, this is not an issue since the measure of influence is always 0 or near 0 and is not part of any parsimonious set and therefore not included in the analysis. However, in one-level analysis, this can cause problems as there is no direction for normal programming. Examining the data, this issue only affects Firm 4 DMC in a one-level analysis. While it is not the only variable with a firm value equal to the industry average (e.g. for firm 2 OPC and OFA have this as well), it is the only variable with a defined formula.

Next the following input is examined; the threshold. While it is already a part of the one and multi level analysis this paragraph focuses on what happens to the explanatory table when the threshold is changed. in figure 18 the second explanatory table shows that PM has 2 contributing and 2 counteracting causes with a threshold of 0.85 this is because variables AC (0.6) and RDC (0.4) are not in the counteracting set since they don't meet the 0.85 threshold, similar to PC (0.31) and DMC (0.69) which are not individually in the contributing set.

When the threshold is changed to 0.5 (Firm 3, Variable = PM, T=0.6, Direction = high, Dimension = 1), the table updates as in figure 21, where AC (0.6) and DMC (0.69) exceed the threshold and explain the high PM. When the threshold is changed to 0.4 while keeping the other inputs the same, the table updates to 22, demonstrating that the tables changes as the threshold changes.

Variable Explained: PM

Firm	Variable	Industry Average	Firm Value	Measure of Influence	Contributing/Counteracting
Firm 3	PM	0.120	0.151		
Firm 3	AC	0.079	0.082	-0.003	Counteracting
Firm 3	DMC	0.072	0.047	0.025	Contributing
Firm 3	RDC	0.007	0.009	-0.002	Counteracting

Figure 22: Python Explanatory Table for firm 3, Variable = PM, T=0.4, Direction = high, Dimension =1

Variable Explained: PM

Firm	Variable	Industry Average	Firm Value	Measure of Influence	Contributing/Counteracting
Firm 3	PM	0.120	0.151		
Firm 3	AC	0.079	0.082	-0.003	Counteracting
Firm 3	DMC	0.072	0.047	0.025	Contributing

Figure 21: Python Explanatory Table for firm 3, Variable = PM, T=0.5, Direction = high, Dimension = 1

Lastly, the direction is examined. Although direction was already part of the one-level and multi-level analyses, this paragraph focuses on the consequences of setting the wrong direction. For instance, if a firm's value is higher than the industry value but the direction is set to low. Consider the ROA of firm 1, which is 0.251 compared to the industry average of 0.137, indicating a high value. However, if the direction is incorrectly set to low, only the ROA table appears, even with a dimension of 5, as illustrated in figure 23. This occurs because when the direction is set opposite to the actual situation, the variables influencing the selected variable will quickly be identified as counteracting, causing the code to stop further analysis.

Variable Explained: ROA

Firm	Variable	Industry Average	Firm Value	Measure of Influence	Contributing/Counteracting
Firm 1	ROA	0.137	0.251		
Firm 1	PM	0.120	0.190	-0.0796	Counteracting
Firm 1	TA	1.140	1.320	-0.0214	Counteracting

Figure 23: Python Explanatory Table for firm 1, Variable = ROA, T=0.85, Direction = low, Dimension = 5

6.2.2 Explanatory Query Output

In this section the output of explanation formalism; the explanatory tree and the output of the ETG; a textual explanation are examined. Just as in the multiple level analysis of the Explanatory Query input the trees and text for the ROA with a dimension of 5 and a threshold of 0.85 was determined for each firm. In figure 24 is the explanatory text generated by the ETG in figure 25 is the explanatory tree drawn from the calculated measures in Appendix E.

The Factors on Why the Return on Assets of Firm 2 is High

Return on Assets

The Return on Assets (ROA) for Firm 2 is significantly high, primarily driven by its *Profit Margin (PM)*. With Firm 2's PM being substantially higher than the industry average, it is strongly *contributing* to the firm's overall ROA. This suggests that Firm 2 is efficient in converting sales into actual profit, which directly boosts its ROA. No *counteracting* factors are noted for ROA, highlighting that the firm's profitability is a major strength.

Profit Margin

Firm 2's Profit Margin (PM) stands out as notably high compared to the industry average. This impressive PM is supported by relatively low *Administrative Costs (AC)* and *Production Cost (PC)*. Firm 2's AC indicates efficient asset management, while its strong PC suggests effective control over product costs and pricing *contributing* to a high PM. However, these positive contributions are somewhat offset by a higher-than-average *Distribution and Marketing Cost (DMC)*, which exerts a *counteracting* influence on the PM. Despite this, the overall effect remains positive, leading to a robust profit margin for the firm.

Production Cost

Firm 2's Production Cost (PC) is above the industry average, bolstered by low *Material Cost (MC)* and *Works Labour Costs (WLC)*. Effective MC indicates Firm 2's proficiency in managing manufacturing processes, while the *contributing* WLC reflects well-managed labor costs. These factors contribute significantly to Firm 2's low PC, ensuring better cost management and pricing strategies.

Material Cost

Material Cost (MC) for Firm 2 is below the industry benchmark, driven primarily by *Metal Used (MU)*. The firm's MU demonstrates efficient use of manufacturing resources, significantly *contributing* its high ROA. However, there is a *counteracting* impact from a marginally higher-than-average *Bought-out Components (BC)*. Despite this minor counteraction, the strong MU ensures that Firm 2 maintains control over its manufacturing processes.

Figure 24: Explanatory Text Firm 2, Variable = ROA, T=0.85, Direction = high, Dimension = 5

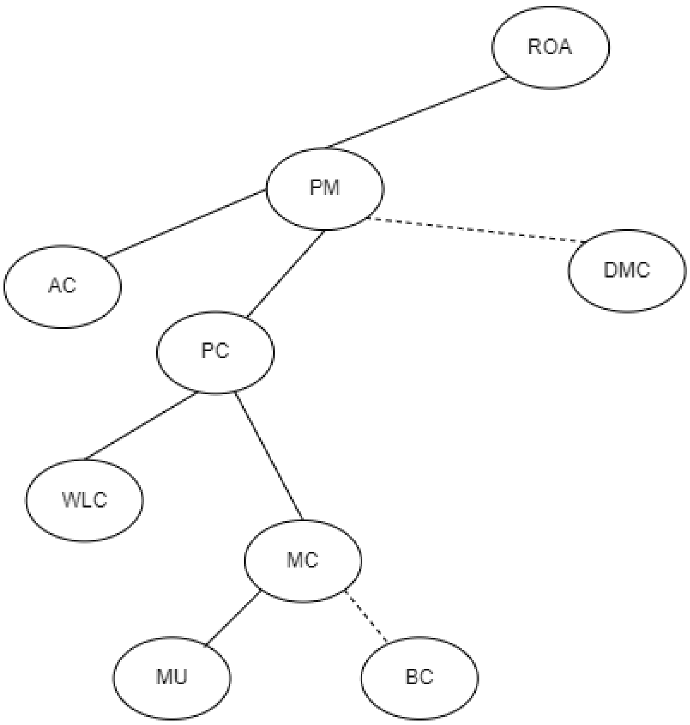


Figure 25: Explanatory Tree Firm 2, Variable = ROA, T=0.85, Direction = high, Dimension = 5

The outputs were assessed by verifying if the variables identified in the tree were also present in the text and whether the same contributing and counteracting variables were recognised. If the tree displayed a dotted line, the text needed to indicate it as counteracting. For all firms, the text consistently identified the same variables, including both contributing and counteracting factors. On one occasion, the text used an incorrect abbreviation for Manufacturing Utilisation (MU). Attempts were made to replicate this error, but they were unsuccessful.

6.3 Operational Comparison

In this section the explanatory tree and text are evaluated as if they were part of a business dashboard by discussing their advantages and disadvantages, also if they were combined. It is important to note that this is hypothetical and should be investigated in a real-world setting to confirm the validity.

Explanatory Tree

The explanation tree has several advantages. It provides visual clarity by offering a clear visual representation of the relationships between different factors, helping users quickly understand the hierarchy and connections. It allows for quick insights, enabling users to see the main contributing and counteracting factors without having to read through a lot of text. Additionally, the explanation tree has interactive potential; in a dashboard, it can be made interactive, allowing users to click on different nodes to get more detailed information.

However, the explanation tree also has some disadvantages. It may not provide the depth of information that some users need, as it offers a high-level overview but may lack detailed explanations. Users might need to interpret the visual elements, which could be challenging if they are not familiar with the format or the specific business context. Some interactions can be counter intuitive; for example, a contributing factor might seem to always have a positive relationship, but there can also be negative relationships.

Explanatory Text

The explanatory text has its own set of advantages. It provides detailed information, offering comprehensive explanations and context for each factor, which is beneficial for in-depth understanding. It ensures clarity in communication by explicitly stating the relationships and impacts of different factors, reducing the need for interpretation. Moreover, text allows for a more nuanced explanation, including subtleties and specific details that might be lost in a visual representation.

On the other hand, the explanatory text can be time-consuming to read, especially in a business dashboard designed for quick insights. A large amount of text can be overwhelming and might lead to information overload for users looking for quick answers.

Combining Both Approaches

Combining both the explanation tree and explanatory text offers several advantages. It provides a comprehensive understanding by catering to both quick insights through the tree and detailed explanations through the text. This approach accommodates different user preferences, allowing users to choose the format that works best for them.

However, there are also disadvantages to combining both methods. It can make the dashboard more complex and potentially harder to navigate. Ensuring consistency between the visual and textual explanations can be challenging and requires careful management.

To conclude, in the context of a business dashboard, the choice between an explanation tree and explanatory text depends on the specific needs and preferences of the users. An explanation tree is ideal for quick insights and visual learners, while explanatory text is more suitable for those who require detailed, contextual information. Combining both can offer a balanced approach, particularly when the text is optional to ensure quick insights and provide comprehensive information when needed. However, this approach requires careful design to maintain usability and coherence.

7 Discussion, Conclusion, Limitations and Future Research

This chapter begins with a discussion of the individual chapters, followed by the conclusion. The limitations section addresses the constraints encountered during the study. Finally, potential directions for future research are explored.

7.1 Discussion

This thesis aimed to enhance business dashboard by combining the power of AI and explanatory analytics, with four research questions to achieve this goal.

RQ1. What are business dashboards, explanatory analytics, prompt engineering, and what do they entail?

Business dashboards are interactive tools that visually present key metrics to aid in the recognition and diagnosis phase of the decision-making process. They integrate various data sources to offer a comprehensive view of business performance. Explanatory analytics extends this capability by providing insights into the underlying causes of observed data patterns. Four models for explanations, namely explanation formalism (Feelders and Daniels, 2001), informative summarization (Sarawagi, 2001), explanation by intervention (Roy and Suciu, 2014), and GPT-4 are explained thoroughly and compared in section 1. Their algorithms vary: top-down, bottom-up, big bang, and transformer-based neural network, respectively. They also differ in how they derive actual and reference values, with complex formulas used in the first three methods and context-driven generation in GPT-4. Explanation formalism was chosen because the input and output could be closely monitored, which is important since AI is used as a black box. Computational methods for these techniques are detailed, showing distinct mathematical models for each approach. Filtering methods include fraction-based filtering, error summarization, and intervention ordering, while GPT-4 uses prompt restrictions. Filtering GPT-4 effectively relies on prompt engineering, which involves crafting specific input prompts to guide AI models in producing relevant and accurate outputs. The literature review highlights the importance of designing clear and unbiased prompts, which is crucial for leveraging AI to generate precise textual explanations.

RQ2. What are the requirements for an artefact that uses AI as a tool for explanatory analytics?

The artefact requires a robust framework integrating AI to enhance its capabilities. Key requirements include the ability to process and analyse data, generate coherent and contextually relevant textual explanations, and present these explanations in an easily interpretable format.

Additionally, the system must ensure accuracy and relevance in the generated explanations. To ensure a comprehensive design that meets these requirements, UML diagrams were created to map out the system's architecture, providing a clear blueprint for implementation.

The framework must handle data correctly, ensuring it is processed efficiently to extract meaningful insights. These insights are crucial for generating explanations that are not only accurate but also relevant to the context. The presentation of these explanations must be clear and easily understandable to ensure that users can readily interpret and utilise the information provided.

Furthermore, the system's architecture, as outlined in the UML diagrams, details the interaction between various components, highlighting how data flows through the system and is transformed into useful explanations. This visual representation aids in identifying potential challenges and optimising the design before development.

RQ3. How to develop an AI-powered artefact for explanatory analytics?

The development of the artefact involves creating and implementing two types of queries within the Explanatory Text Generator (ETG): Descriptive Query and Explanatory Query. Each query type is supported by two dedicated modules—one handling the graphical user interface (GUI) and the other focusing on the back-end logic for generating responses. These modules ensure seamless interaction between users and the system, facilitating accurate responses.

The most important part of this process is the building of the prompt. The prompt construction is critical because it dictates the quality and relevance of the generated responses. Key components of the prompt include the context of the query, the specific details required, and the intended use of the information. The order in which these components are arranged also plays a significant role in ensuring clarity and coherence. First, the context sets the stage by providing background information. Next, the specific data narrow down the focus, guiding the system on what precise information to generate. By carefully structuring these components, the system can produce accurate explanations.

RQ4. How to evaluate the addition of AI to explanatory analytics within a business dashboard environment?

Evaluation involves comparing the AI-enhanced explanatory analytics with traditional methods by using a a cognitive walkthrough to evaluate the usability and user experience of the AI-powered artefact. This involves simulating a user's problem-solving process and identifying potential usability issues. This test assessed how well the artefact performed compared to traditional explanatory analytics tools.

7.2 Conclusion

Integrating AI into business dashboards significantly enhances their explanatory analytics capabilities. AI can handle descriptive queries on its own, making it a versatile tool for straightforward information needs. For more detailed explanations, it requires input from the explanation formalism algorithm.

By combining existing explanatory models with GPT-4, AI can generate coherent and contextually relevant textual explanations instead of figures, particularly helpful when a user is lacking data analysis skills or domain knowledge. This aids in understanding complex data patterns, facilitating more informed decision-making, and bridging the gap between data analysis and insights. The developed artefact showcases AI's potential to assist business intelligence by providing more intuitive explanations.

7.3 Limitations

The study has several limitations. Firstly, the system does not include a classifier to determine whether a query is descriptive or explanatory. As a result, the artefact was not an integrated solution but rather comprised two separate modules. Secondly, the explanatory query artefact lacks a mechanism for dealing with situations when the actual value is equal to the reference value. Thirdly, the artefact does not effectively address qualitative variables, particularly MU and SPP, since these variables have no intrinsic value. These limitations could result in a less comprehensive understanding of the underlying causes.

Furthermore, this artefact was designed based on the firm's value and industry average. While this approach is theoretically sound, in practice, the industry average is not always readily available. Therefore, using a time reference value, such as years or quarters, would be more practical and effective.

Additionally, the artefact is a prototype and has not been tested extensively in real-world business environments, limiting its applicability and scalability. The evaluations were conducted in controlled settings, which do not fully represent real-world complexities and variabilities, thus affecting the generalisability of the findings. Furthermore, GPT-4 was the only AI model used in this study, and no other AI models were tested, which may limit the understanding of how different models could perform under similar conditions.

7.4 Future Research

Future research should focus on several areas to enhance the current artefact. Developing a classifier to better manage the queries would improve the system's effectiveness in creating one integrated system. Expanding the system to handle qualitative relations more effectively would provide a more comprehensive analysis of business scenarios, capturing the full context and underlying causes. Conducting real-world testing is essential to validate the artefact's performance and adaptability in diverse business environments, ensuring its robustness and scalability.

Additionally, the necessity of integrating textual explanations as discussed in section operational comparison should be tested by users. It might turn out that users prefer visual explanations.

Moreover, as AI continues to evolve, and GPT shows potential in handling specific areas of expertise such as business intelligence, data analysis, or economics, it would be interesting to explore building AI models with domain-specific explanations. Such advancements could enhance the precision and relevance of AI-generated insights, potentially removing the need for underlying algorithms like the explanation formalism altogether. This approach could lead to more effective integration of AI in the explanatory analytics domain.

In conclusion, while the current artefact shows promising results in enhancing explanatory analytics with AI, further research and development are needed to address its limitations and fully realise its potential in real-world applications.

References

Akkio. Akkio. https://akkio.com, 2024. Accessed: 2024-05-28.

Altexsoft. Complete guide to business intelligence and analytics: Strategy, steps, processes, and toolse. https://www.altexsoft.com/blog/complete-guide-to-business-intelligence-and-analytics-strategy-steps-processes-and-tools/, 2023.

António Lorvão Antunes, Elsa Cardoso, and José Barateiro. Incorporation of ontologies in data warehouse/business intelligence systems-a systematic literature review. *International Journal of Information Management Data Insights*, 2(2):100131, 2022.

Aaro Askala. Explanatory analytics in business dashboards. *Turku University Thesis*, 2020.

Ranjit Bose. Advanced analytics: opportunities and challenges. *Industrial Management & Data Systems*, 109(2):155–172, 2009.

Emma Bowman. A new ai chatbot might do your homework for you. but it's still not an a+ student. *NRP*, 2022.

Katerina Božič and Vlado Dimovski. Business intelligence and analytics use, innovation ambidexterity, and firm performance: A dynamic capabilities perspective. *The Journal of Strategic Information Systems*, 28(4):101578, 2019. ISSN 0963-8687. doi: https://doi.org/10.1016/j.jsis.2019.101578.

Erik Brynjolfsson, Danielle Li, and Lindsey R Raymond. Generative ai at work. Technical report, National Bureau of Economic Research, 2023.

Erik Cambria and Bebo White. Jumping nlp curves: A review of natural language processing research. *IEEE Computational intelligence magazine*, 9(2):48–57, 2014.

Capterra. New to dashboard? get familiar with benefits, features & more. https://www.capterra.com/dashboard-software/, 2024. Accessed: 2024-05-28.

E. Caron. Explanation of exceptional values in multi-dimensional business databases. *PhD thesis, Erasmus Univeristy Rotterdam*, 2013.

ChatPowerBI. Chatpowerbi. https://chatpowerbi.ai, 2024. Accessed: 2024-05-28.

Surajit Chaudhuri and Umeshwar Dayal. An overview of data warehousing and olap technology. *ACM Sigmod record*, 26(1):65–74, 1997.

Surajit Chaudhuri, Umeshwar Dayal, and Vivek Narasayya. An overview of business intelligence technology. *Communications of the ACM*, 54(8):88–98, 2011.

Hsinchun Chen, Roger H. L. Chiang, and Veda C. Storey. Business intelligence and analytics: From big data to big impact. *MIS Quarterly*, 36(4):1165–1188, 2012. ISSN 02767783. URL http://www.jstor.org/stable/41703503.

Rene Y Choi, Aaron S Coyner, Jayashree Kalpathy-Cramer, Michael F Chiang, and J Peter Campbell. Introduction to machine learning, neural networks, and deep learning. *Translational vision science & technology*, 9(2):14–14, 2020.

Consensus. Ai search engine for research. https://consensus.app/, 2024. Accessed: 2024-27-07.

Benoit Dageville, Thierry Cruanes, Marcin Zukowski, Vadim Antonov, Artin Avanes, Jon Bock, Jonathan Claybaugh, Daniel Engovatov, Martin Hentschel, Jiansheng Huang, et al. The snowflake elastic data warehouse. In *Proceedings of the 2016 International Conference on Management of Data*, pages 215–226, 2016.

Ali Darejeh and Dalbir Singh. A review on user interface design principles to increase software usability for users with less computer literacy. *Journal of computer science*, 9(11):1443, 2013.

Datamatics. Book a demo - intelligent automation. https://datamatics.com/intelligent-automation/book-a-demo, 2024. Accessed: 2024-05-28.

Behrooz Davazdahemami, Hamed M Zolbanin, and Dursun Delen. An explanatory analytics framework for early detection of chronic risk factors in pandemics. *Healthcare Analytics*, 2: 100020, 2022.

Jianyang Deng and Yijia Lin. The benefits and challenges of chatgpt: An overview. *Frontiers in Computing and Intelligent Systems*, 2(2):81–83, 2022.

Wayne W Eckerson. *Performance dashboards: measuring, monitoring, and managing your business*. John Wiley & Sons, 2010.

Elicit. Analyze research papers. https://elicit.com/?redirected=true, 2024. Accessed: 2024-27-07.

A.J. Feelders and H.A.M. Daniels. A general model for automated business diagnosis. *European Journal of Operational Research*, 130(3):623–637, 2001. ISSN 0377-2217. doi: https://doi.org/10.1016/S0377-2217(99)00428-2.

Éric Foley and Manon G Guillemette. What is business intelligence? *International Journal of Business Intelligence Research (IJBIR)*, 1(4):1–28, 2010.

Michael Friedman. Explanation and scientific understanding. *The Journal of Philosophy*, 71 (1):5–19, 1974. ISSN 0022362X. URL http://www.jstor.org/stable/2024924.

Geckoboard. Dashboards examples. https://www.geckoboard.com/dashboard-examples/, 2024.

Judy Wawira Gichoya, Kaesha Thomas, Leo Anthony Celi, Nabile Safdar, Imon Banerjee, John D Banja, Laleh Seyyed-Kalantari, Hari Trivedi, and Saptarshi Purkayastha. Ai pitfalls and what not to do: mitigating bias in ai. *The British Journal of Radiology*, 96(1150): 20230023, 2023.

Github. Pypl popularity of programming language. https://pypl.github.io/PYPL.html, 2024. Accessed: 2024-04-03.

Matteo Golfarelli, Stefano Rizzi, and Iuris Cella. Beyond data warehousing: what's next in business intelligence? *ACM Seventh International Workshop on Data Warehousing and OLAP*, pages 1–6, 11 2004. doi: 10.1145/1031763.1031765.

Jiawei Han, Micheline Kamber, and Jian Pei. Data mining: Concepts and techniques. *Elsevier*, 2011.

Alan R Hevner, Salvatore T March, Jinsoo Park, and Sudha Ram. Design science in information systems research. *MIS quarterly*, pages 75–105, 2004.

Andreas Holzinger, Georg Langs, Helmut Denk, Kurt Zatloukal, and Heimo Müller. Causability and explainability of artificial intelligence in medicine. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 9(4):e1312, 2019.

Involve. Ai powered analytics. https://involve.me/ai-powered-analytics, 2024. Accessed: 2024-05-28.

Oyku Isik, Mary Jones, and Anna Sidorova. Business intelligence success: The roles of bi capabilities and decision environments. *Information 'I&' Management*, 50:13–23, 12 2012. doi: 10.1016/j.im.2012.12.001.

Dietmar Jannach, Ahtsham Manzoor, Wanling Cai, and Li Chen. A survey on conversational recommender systems. *ACM Computing Surveys (CSUR)*, 54(5):1–36, 2021.

Kshiti D Joshi, Lei Chi, Avimanyu Datta, and Shu Han. Changing the competitive landscape: Continuous innovation through it-enabled knowledge capabilities. *Information Systems Research*, 21(3):472–495, 2010.

Zack Jourdan, R Kelly Rainer, and Thomas E Marshall. Business intelligence: An analysis of the literature. *Information systems management*, 25(2):121–131, 2008.

Krishna Prakash Kalyanathaya, D Akila, and P Rajesh. Advances in natural language processing–a survey of current research trends, development tools and industry applications. *International Journal of Recent Technology and Engineering*, 7(5C):199–202, 2019.

Yue Kang, Zhao Cai, Chee-Wee Tan, Qian Huang, and Hefu Liu. Natural language processing (nlp) in management research: A literature review. *Journal of Management Analytics*, 7(2): 139–172, 2020.

Triin Kask. Innovation as a result of strategic decisions in the context of organizational environment: the case of estonian information and communication technology companies. *tartu Ülikool*, 2019.

Vishnu Singh Khatuwal and Digvijay Puri. Business intelligence tools for dashboard development. In *2022 3rd International Conference on Intelligent Engineering and Management (ICIEM)*, pages 128–131, 2022. doi: 10.1109/ICIEM54221.2022.9853086.

Ralph Kimball and Margy Ross. *The data warehouse toolkit: the complete guide to dimensional modeling*. John Wiley & Sons, 2011.

Mark Levene and George Loizou. Why is the snowflake schema a good data warehouse design? *Information Systems*, 28(3):225–240, 2003.

Leszek A Maciaszek. *Requirements analysis and system design: developing information systems with UML*. Addison-Wesley Longman Ltd., 2001.

Thomas Mahatody, Mouldi Sagar, and Christophe Kolski. State of the art on the cognitive walkthrough method, its variants and evolutions. *Intl. Journal of Human–Computer Interaction*, 26(8):741–785, 2010.

Timothy R McIntosh, Tong Liu, Teo Susnjak, Paul Watters, Alex Ng, and Malka N Halgamuge. A culturally sensitive test to evaluate nuanced gpt hallucination. *IEEE Transactions on Artificial Intelligence*, 2023.

Microsoft. Openai service. https://azure.microsoft.com/nl-nl/products/ai-services/openai-service, 2024a. Accessed: 2024-05-28.

Microsoft. Powerbi website. https://powerbi.microsoft.com/, 2024b. Accessed: 2024-05-21.

Henry Mintzberg. *Mintzberg on management: Inside our strange world of organizations*. Simon and Schuster, 1989.

Grégoire Montavon, Wojciech Samek, and Klaus-Robert Müller. Methods for interpreting and understanding deep neural networks. *Digital Signal Processing*, 73:1–15, 2018. ISSN 1051-2004. doi: https://doi.org/10.1016/j.dsp.2017.10.011.

Abdallah Moubayed, MohammadNoor Injadat, Ali Bou Nassif, Hanan Lutfiyya, and Abdallah Shami. E-learning: Challenges and research opportunities using machine learning & data analytics. *IEEE Access*, 6:39117–39138, 2018. URL `https://api.semanticscholar.org/CorpusID:51921148`.

Solomon Negash. Business intelligence. *Communications of the association for information systems*, 13(1):15, 2004.

Solomon Negash and Paul Gray. Business intelligence. *Handbook on decision support systems 2*, pages 175–193, 2008.

OpenAI. Openai. https://www.openai.com, 2024.

Overleaf. Use grammarly with overleaf. https://www.overleaf.com/learn/how-to/Use_Grammarly_with_Overleaf, 2024. Accessed: 2024-27-07.

OverleafGPT. nteractive assistant for writing detailed overleaf documents. https://chatgpt.com/g/g-RAgF4tibg-overleaf-gpt, 2024. Accessed: 2024-27-07.

Stefan Pauliuk. Critical appraisal of the circular economy standard bs 8001:2017 and a dashboard of quantitative system indicators for its implementation in organizations. *Resources, Conservation and Recycling*, 129:81 – 92, 2018. doi: 10.1016/j.resconrec.2017.10.019. URL `https://www.scopus.com/inward/record.uri?eid=2-s2.0-85032859874&doi=10.1016%2fj.resconrec.2017.10.019&partnerID=40&md5=22ef01555b40dc08aef60c322f176ded`. Cited by: 350; All Open Access, Hybrid Gold Open Access.

Fernando Perez, Brian E Granger, and John D Hunter. Python: an ecosystem for scientific computing. *Computing in Science & Engineering*, 13(2):13–21, 2010.

Plotly. Get a demo. https://plotly.com/get-demo/, 2024. Accessed: 2024-05-28.

Polymer. Ai dashboard generator. https://www.polymersearch.com/ai-dashboard-generator, 2024. Accessed: 2024-05-28.

PowerBI. Power bi. https://app.powerbi.com/, 2024. Accessed: 2024-05-28.

PythongGPT. a highly sophisticated gpt tailored for python programmers. https://chatgpt.com/g/g-cKXjWStaE-python, 2024. Accessed: 2024-27-07.

Nils H Rasmussen, Manish Bansal, and Claire Y Chen. *Business dashboards: a visual catalog for design and deployment*. John Wiley 'I&' Sons, 2009.

Sudeepa Roy and Dan Suciu. A formal approach to finding explanations for database queries. In *Proceedings of the 2014 ACM SIGMOD international conference on Management of data*, pages 1579–1590, 2014.

Thammasak Rujirayanyong and Jonathan Shi. A project-oriented data warehouse for construction. *Automation in Construction*, 15:800–807, 11 2006. doi: 10.1016/j.autcon.2005.11.001.

Sunita Sarawagi. idiff: Informative summarization of differences in multidimensional aggregates. *Data Mining and Knowledge Discovery*, 5:255–276, 2001.

Jacco Willem Spruijt. Explanatory analytics for business dashboard – improving decision-making through augmented dashboards –. *Tilburg University Master Thesis*, 2023.

statistica. Chatgpt marketshare. *https://www.statista.com/chat/31199/share-of-worldwide-ai-text-generation-tool-users/*, 2024.

Tableau. Tableau ai. https://www.tableau.com/products/tableau-ai, 2024. Accessed: 2024-05-28.

Salwa Tanko, Hawra Al-Saeed, Ruqaya Alali, Sara Alawami, Samiha Brahimi, and Gomathi Kirishina. Performance dashboard for productive families. In *2019 2nd International Conference on Computer Applications 'I&' Information Security (ICCAIS)*, pages 1–4, 2019. doi: 10.1109/CAIS.2019.8769481.

Paul Thagard. Can chatgpt make explanatory inferences? benchmarks for abductive reasoning. *ArXiv*, abs/2404.18982, 2024. URL `https://api.semanticscholar.org/CorpusID:269457635`.

Ashish Thusoo, Zheng Shao, Suresh Anthony, Dhruba Borthakur, Namit Jain, Joydeep Sen Sarma, Raghotham Murthy, and Hao Liu. Data warehousing and analytics infrastructure at facebook. In *Proceedings of the ACM SIGMOD International Conference on Management of Data*, pages 1013–1020, 2010. doi: 10.1145/1807167.1807278. Cited by: 309.

Nectaria Tryfona, Frank Busborg, and Jens G Borch Christiansen. starer: A conceptual model for data warehouse design. In *Proceedings of the 2nd ACM international workshop on Data warehousing and OLAP*, pages 3–8, 1999.

Charles Tupper. *Data architecture: from zen to reality*. Elsevier, 2011.

IBM Turbonomic. Dashboards. https://www.ibm.com/products/turbonomic/dashboards, 2024. Accessed: 2024-05-28.

Stijn Viaene. Data scientists aren't domain experts. *IT Professional*, 15:12–17, 2013. URL `https://api.semanticscholar.org/CorpusID:10361021`.

Claire Vink. Automated diagnostic analytics in databases and business dashboards. *Tilburg University Bachelor Thesis*, 2020.

Conny Walchshofer, Vaishal Dhanoa, Marc Streit, and Miriah Meyer. Transitioning to a commercial dashboarding system: Socio-technical observations and opportunities. *IEEE Transactions on Visualization and Computer Graphics*, 2023.

Zirui Wang, Jun Wang, and Youren Wang. An intelligent diagnosis scheme based on generative adversarial learning deep neural networks and its application to planetary gearbox fault pattern recognition. *Neurocomputing*, 310:213–222, 2018.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022.

Jules White, Quchen Fu, Sam Hays, Michael Sandborn, Carlos Olea, Henry Gilbert, Ashraf Elnashar, Jesse Spencer-Smith, and Douglas C Schmidt. A prompt pattern catalog to enhance prompt engineering with chatgpt. *arXiv preprint arXiv:2302.11382*, 2023.

Yue Zhang, Yafu Li, Leyang Cui, Deng Cai, Lemao Liu, Tingchen Fu, Xinting Huang, Enbo Zhao, Yu Zhang, Yulong Chen, et al. Siren's song in the ai ocean: a survey on hallucination in large language models. *arXiv preprint arXiv:2309.01219*, 2023.

Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Claire Cui, Olivier Bousquet, Quoc Le, et al. Least-to-most prompting enables complex reasoning in large language models. *arXiv preprint arXiv:2205.10625*, 2022.

Appendices

Appendix A

Variables	Firm 1	Firm 2	Firm 3	Firm 4	Firm 5	Firm 6	Firm 7	Firm 8	Firm 9	AVG
ROA	0.251	0.239	0.189	0.161	0.133	0.132	0.088	0.079	0.041	0.137
PM	0.19	0.199	0.151	0.099	0.103	0.115	0.087	0.089	0.047	0.12
TA	1.32	1.2	1.25	1.63	1.29	1.15	1.01	0.9	0.87	1.14
OA	0.758	0.833	0.8	0.77	0.869	0.809	0.754	0.774	0.804	0.877
PC	0.628	0.635	0.711	0.747	0.725	0.719	0.754	0.774	0.804	0.722
RDC	0.005	0.01	0.009	0.007	0.007	0.011	0.014	0	0.002	0.007
DMC	0.109	0.116	0.047	0.072	0.062	0.058	0.073	0.046	0.064	0.072
AC	0.068	0.04	0.082	0.075	0.103	0.097	0.072	0.091	0.085	0.079
MC	0.32	0.287	0.339	0.301	0.397	0.316	0.336	0.347	0.358	0.333
WLC	0.165	0.221	0.232	0.283	0.157	0.241	0.248	0.274	0.289	0.234
OPC	0.143	0.127	0.14	0.092	0.104	0.113	0.114	0.153	0.155	0.127
WS	0	0	0	0.071	0.067	0.049	0.056	0	0	0.027
CA	0.465	0.481	0.412	0.35	0.369	0.449	0.549	0.582	0.608	0.474
FA	0.293	0.352	0.388	0.263	0.406	0.42	0.441	0.529	0.541	0.404
MS	0.08	0.11	0.081	0.079	0.068	0.092	0.1	0.101	0.082	0.088
WIP	0.043	0.04	0.063	0.062	0.083	0.106	0.188	0.225	0.245	0.117
FGS	0.132	0.102	0.047	0.037	0.039	0.045	0.044	0.053	0.057	0.062
DEB	0.21	0.229	0.221	0.172	0.179	0.206	0.217	0.203	0.224	0.207
LB	0.13	0.158	0.194	0.169	0.208	0.214	0.311	0.277	0.268	0.214
PAM	0.16	0.189	0.19	0.089	0.194	0.203	0.123	0.246	0.264	0.184
OFA	0.003	0.005	0.004	0.005	0.004	0.003	0.007	0.006	0.009	0.005
EX	0.16	0.4	0.15	0	0.25	0.23	0.1	0.12	0.3	0.19
SPP	2	2	1	1	1	1	1	1	1	1.22
BC	0.32	0.39	0.29	0.33	0.56	0.35	0.4	0.37	0.43	0.38
MU	2	1	2	1	2	1	1	2	1	1.44

ROA = PM * TA

PM = 1 - (AC + PC + RDC + DMC)

TA = 1 / OA

OA = CA + FA

PC = MC + WLC + WS + OPC

CA = FGS + DEB + MS + WIP

FA = LB + PAM + OFA

WIP = ’SPP’: ’neg’

MC = ’MU’: ’pos’, ’BC’: ’pos’

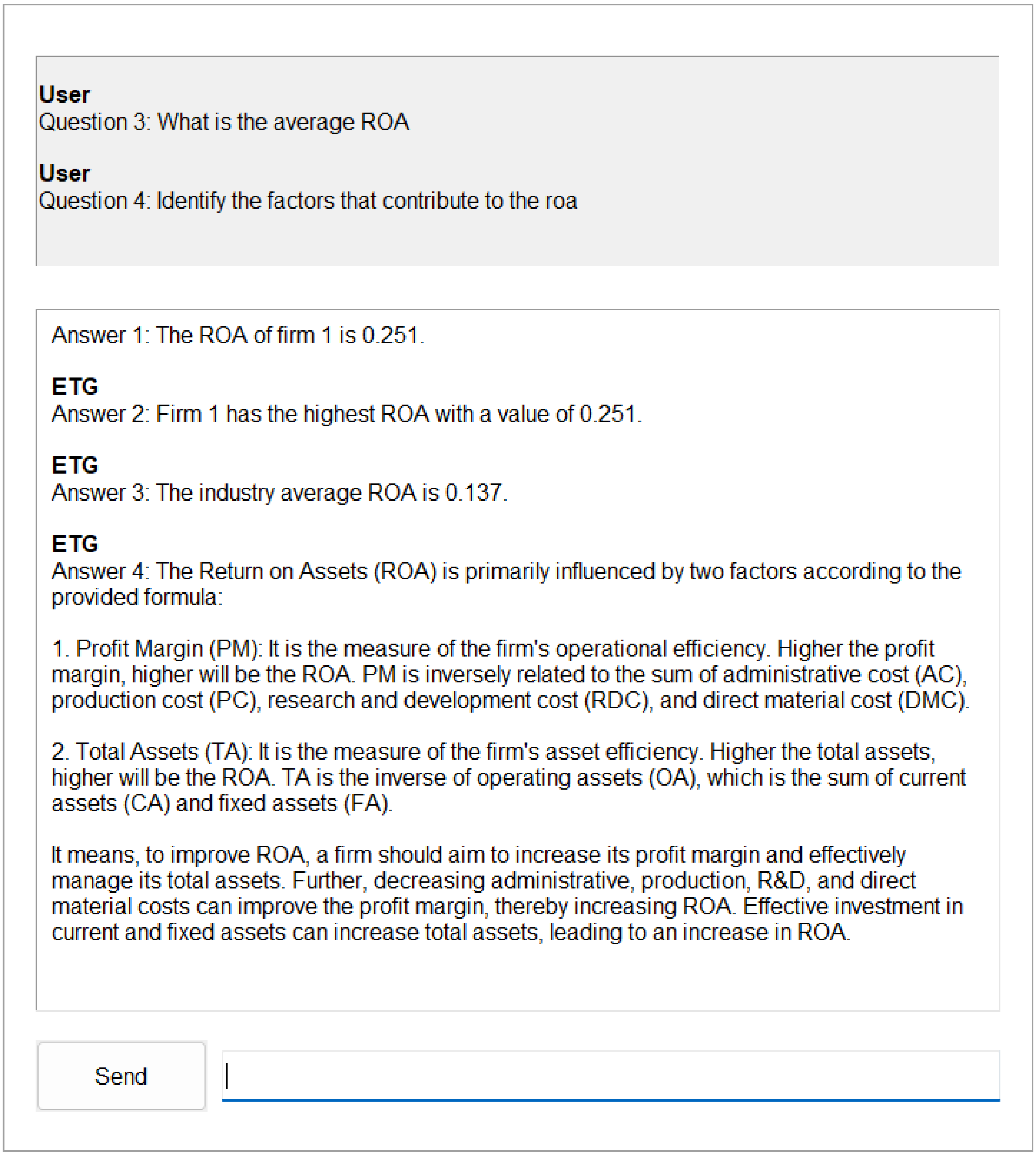
DMC = ’SPP’: ’pos’, ’EX’: ’pos’

FGS = ’SPP’: ’pos’

DEB = ’EX’: ’pos’

(Feelders and Daniels, 2001)

Appendix B



B1. Interface Descriptive Query

Appendix C

Descriptive Query

```python
import pandas as pd
import openai

openai.api_key = key used in project is removed for privacy and fraud prevention reasons
class DataAnalysis:
    def __init__(self):
        self.fixed_file_path = r"C:\Users\jeroe\Code\Data Firm Analysis revised.xlsx"
        self.formulas = """
        ROA = PM * TA
        PM = 1 - (AC + PC + RDC + DMC)
        TA = 1 / OA
        OA = CA + FA
        PC = MC + WLC + WS + OPC
        CA = FGS + DEB + MS + WIP
        FA = LB + PAM + OFA
        WIP = {'SPP': 'neg'}
        MC = {'MU': 'pos', 'BC': 'pos'}
        DMC = {'SPP': 'pos', 'EX': 'pos'}
        FGS = {'SPP': 'pos'}
        DEB = {'EX': 'pos'}
        """
        self.conversation_history = []
        self.load_data()

    def load_data(self):
        try:
            self.df = pd.read_excel(self.fixed_file_path)
            print("Data loaded successfully.")
        except Exception as e:
            print(f"An error occurred: {str(e)}")

    def send_chat(self, user_message):
        if user_message:
            try:
                self.conversation_history.append({"role": "user", "content": user_message})

                # Add the Excel data and formulas to the conversation history only once
                if not any("Excel Data:" in message["content"] for message in self.conversation_history):
                    data_prompt = self.df.to_string(index=False)
```

```python
                analysis_prompt = f"You are given the following data. Use only the provided data and the formulas, it is important to give only the information that is asked to keep the answer as short as possible. :\n{data_prompt}\n\n{self.formulas}"
                self.conversation_history.append({"role": "system", "content": analysis_prompt})

            response = openai.ChatCompletion.create(
                model="gpt-4",
                messages=self.conversation_history,
                max_tokens=1500
            )

            response_content = response['choices'][0]['message']['content']
            self.conversation_history.append({"role": "assistant", "content": response_content})
            return response_content
        except Exception as e:
            return f"An error occurred: {str(e)}"
```

Appendix D

Explanation Formalism

```python
import pandas as pd
import itertools

def filter_variables(all_counteracting_variables, fixed_list):
    return [var for var in all_counteracting_variables if var in fixed_list]

def get_missing_variables(filtered_counteracting_variables, filtered_dimensions):
    missing_variables = set()
    for key in filtered_counteracting_variables:
        if key in filtered_dimensions:
            missing_variables.update(filtered_dimensions[key])
    return list(missing_variables)

def calculate_firm_metrics(data, firm_name, selected_variable, direction, threshold):
    firm_values = data.set_index('Variables')[firm_name].to_dict()
    industry_values = data.set_index('Variables')['Industry Average'].to_dict()
    influence_sign = -1 if direction == 'low' else 1

    influences = {}
    if selected_variable == 'ROA':
        influences['Influence_PM_ROA'] = influence_sign * round((firm_values['PM'] * industry_values['TA']) - industry_values['ROA'], 4)
        influences['Influence_TA_ROA'] = influence_sign * round((firm_values['TA'] * industry_values['PM']) - industry_values['ROA'], 4)
    elif selected_variable == 'PM':
        influences['Influence_AC_PM'] = influence_sign * round((1 - (firm_values['AC'] + industry_values['PC'] + industry_values['RDC'] + industry_values['DMC'])) - industry_values['PM'], 4)
        influences['Influence_PC_PM'] = influence_sign * round((1 - (industry_values['AC'] + firm_values['PC'] + industry_values['RDC'] + industry_values['DMC'])) - industry_values['PM'], 4)
        influences['Influence_DMC_PM'] = influence_sign * round((1 - (industry_values['AC'] + industry_values['PC'] + industry_values['RDC'] + firm_values['DMC'])) - industry_values['PM'], 4)
        if firm_values['RDC'] != 0:
            influences['Influence_RDC_PM'] = influence_sign * round((1 - (industry_values['AC'] + industry_values['PC'] + firm_values['RDC'] + industry_values['DMC'])) - industry_values['PM'], 4)
    elif selected_variable == 'TA':
        influences['Influence_OA_TA'] = influence_sign * round((1 / firm_values['OA']) - industry_values['TA'], 4)
    elif selected_variable == 'OA':
        specific_influence_sign = (
            1 if direction == 'high' and selected_variable == 'OA' else
```

```
            -1 if direction == 'low' and dimension == 1 else
            -1 if direction == 'high' and dimension != 1 else
            1
        )
        influences['Influence_CA_OA'] = specific_influence_sign * round(firm_values['CA'] + industry_values['FA'] - industry_values['OA'], 4)
        influences['Influence_FA_OA'] = specific_influence_sign * round(industry_values['CA'] + firm_values['FA'] - industry_values['OA'], 4)
    elif selected_variable == 'PC':
        specific_influence_sign = (
            1 if direction == 'high' and dimension == 1 else
            -1 if direction == 'low' and dimension == 1 else
            -1 if direction == 'high' and dimension != 1 else
            1
        )
        influences['Influence_MC_PC'] = specific_influence_sign * round(firm_values['MC'] + industry_values['WLC'] + industry_values['WS'] + industry_values['OPC'] - industry_values['PC'], 4)
        influences['Influence_WLC_PC'] = specific_influence_sign * round(industry_values['MC'] + firm_values['WLC'] + industry_values['WS'] + industry_values['OPC'] - industry_values['PC'], 4)
        influences['Influence_OPC_PC'] = specific_influence_sign * round(industry_values['MC'] + industry_values['WLC'] + industry_values['WS'] + firm_values['OPC'] - industry_values['PC'], 4)
        if firm_values['WS'] != 0:
            influences['Influence_WS_PC'] = specific_influence_sign * round(industry_values['MC'] + industry_values['WLC'] + firm_values['WS'] + industry_values['OPC'] - industry_values['PC'], 4)
    elif selected_variable == 'CA':
        specific_influence_sign = (
            1 if direction == 'high' and selected_variable == 'CA' or selected_variable == 'OA' else
            -1 if direction == 'low' and dimension == 1 else
            -1 if direction == 'high' and dimension != 1 else
            1
        )
        influences['Influence_FGS_CA'] = specific_influence_sign * round(firm_values['FGS'] + industry_values['DEB'] + industry_values['MS'] + industry_values['WIP'] - industry_values['CA'], 4)
        influences['Influence_DEB_CA'] = specific_influence_sign * round(industry_values['FGS'] + firm_values['DEB'] + industry_values['MS'] + industry_values['WIP'] - industry_values['CA'], 4)
        influences['Influence_MS_CA'] = specific_influence_sign * round(industry_values['FGS'] + industry_values['DEB'] + firm_values['MS'] + industry_values['WIP'] - industry_values['CA'], 4)
```

```python
        influences['Influence_WIP_CA'] = specific_influence_sign * round(industry_values['FGS'] + industry_values['DEB'] + industry_values['MS'] + firm_values['WIP'] - industry_values['CA'], 4)
    elif selected_variable == 'FA':
        specific_influence_sign = (
            1 if direction == 'high' and selected_variable == 'FA' or selected_variable == 'OA' else
            -1 if direction == 'low' and dimension == 1 else
            -1 if direction == 'high' and dimension != 1 else
            1
        )
        influences['Influence_LB_FA'] = specific_influence_sign * round(firm_values['LB'] + industry_values['PAM'] + industry_values['OFA'] - industry_values['FA'], 4)
        influences['Influence_PAM_FA'] = specific_influence_sign * round(industry_values['LB'] + firm_values['PAM'] + industry_values['OFA'] - industry_values['FA'], 4)
        influences['Influence_OFA_FA'] = specific_influence_sign * round(industry_values['LB'] + industry_values['PAM'] + firm_values['OFA'] - industry_values['FA'], 4)
    elif selected_variable == 'WIP':
        specific_influence_sign = (
            1 if direction == 'high' and dimension == 1 else
            -1 if direction == 'low' and dimension == 1 else
            -1 if direction == 'high' and dimension != 1 else
            1
        )
        influences['Influence_SPP_WIP'] = specific_influence_sign * round(((firm_values['SPP']- industry_values['SPP'])*-1), 4)
    elif selected_variable == 'MC':
        specific_influence_sign = (
            1 if direction == 'high' and dimension == 1 else
            -1 if direction == 'low' and dimension == 1 else
            -1 if direction == 'high' and dimension != 1 else
            1
        )
        influences['Influence_MU'] = specific_influence_sign * round((firm_values['MU']- industry_values['MU']), 4)
        influences['Influence_BC'] = specific_influence_sign * round((firm_values['BC'] - industry_values['BC']), 4)
    elif selected_variable == 'DMC':
        specific_influence_sign = (
            1 if direction == 'high' and selected_variable == 1 else
            -1 if direction == 'low' and dimension == 1 else
            -1 if direction == 'high' and dimension != 1 else
            1
```

```
        )
        influences['Influence_SPP_DMC'] = specific_influence_sign * round((firm_values['SPP']- industry_values['SPP']), 4)
        if firm_values['EX'] != 0:
            influences['Influence_EX_DMC'] = specific_influence_sign * round((firm_values['EX'] - industry_values['EX']), 4)
    elif selected_variable == 'FGS':
        specific_influence_sign = (
            1 if direction == 'high' and dimension == 1 else
            -1 if direction == 'low' and dimension == 1 else
            -1 if direction == 'high' and dimension != 1 else
            1
        )
        influences['Influence_SPP_FGS'] = specific_influence_sign * round((firm_values['SPP']- industry_values['SPP']), 4)
    elif selected_variable == 'DEB':
        specific_influence_sign = (
             1 if direction == 'high' and dimension == 1 else
            -1 if direction == 'low' and dimension == 1 else
            -1 if direction == 'high' and dimension != 1 else
            1
        )
        if firm_values['EX'] != 0:
            influences['Influence_SPP_DEB'] = specific_influence_sign * round((firm_values['EX']- industry_values['EX']), 4)

    results = []

    def add_influence_result(variable, industry_value, firm_value, influence):
        return {
            'Firm': firm_name,
            'Variable': variable,
            'Industry Average': industry_value,
            'Firm Value': firm_value,
            'Measure of Influence': round(influence, 4),
            'Contributing/Counteracting': 'Contributing' if influence > 0 else 'Counteracting'
        }

    for influence in influences:
        var_name = influence.split('_')[1]
        results.append(add_influence_result(var_name, industry_values[var_name], firm_values[var_name], influences[influence]))

    df = pd.DataFrame(results)

```

```python
    if df.empty or 'Contributing/Counteracting' not in df.columns:
        return df, pd.DataFrame(), pd.DataFrame()

    def calculate_parsimonious_set(df, threshold, effect_type):
        total_influence = df[df['Contributing/Counteracting'] == effect_type]['Measure of Influence'].sum()
        df['Percentage of Influence'] = df.apply(
            lambda row: round((row['Measure of Influence'] / total_influence), 4) if row['Contributing/Counteracting'] == effect_type else '',
            axis=1
        )

        vars_effect = df[df['Contributing/Counteracting'] == effect_type]
        individual_effect = vars_effect[vars_effect['Percentage of Influence'].astype(float) >= threshold]

        if not individual_effect.empty:
            parsimonious_set = individual_effect
        else:
            parsimonious_set = pd.DataFrame()
            for r in range(2, len(vars_effect) + 1):
                for combo in itertools.combinations(vars_effect.index, r):
                    if vars_effect.loc[list(combo), 'Percentage of Influence'].astype(float).sum() >= threshold:
                        parsimonious_set = vars_effect.loc[list(combo)]
                        break
                if not parsimonious_set.empty:
                    break

        return parsimonious_set

    contributing_parsimonious_set = calculate_parsimonious_set(df, threshold, 'Contributing')
    counteracting_parsimonious_set = calculate_parsimonious_set(df, threshold, 'Counteracting')

    final_set = pd.concat([contributing_parsimonious_set, counteracting_parsimonious_set])
    final_set = final_set.sort_index().reset_index(drop=True)
    final_set = final_set.drop(columns=['Percentage of Influence'])

    explained_variable_row = {
        'Firm': firm_name,
        'Variable': selected_variable,
        'Industry Average': industry_values[selected_variable],
        'Firm Value': firm_values[selected_variable],
```

```
        'Measure of Influence': '',
        'Contributing/Counteracting': ''
    }
    final_set = pd.concat([pd.DataFrame([explained_variable_row]), final_set], ignore_index=True)

    return final_set, contributing_parsimonious_set, counteracting_parsimonious_set

    def generate_report(data, firm_name, dimension, selected_variable, direction, threshold, output_file):

    variables = variables_by_dimension[selected_variable][dimension-1]

    all_counteracting_variables = set()
    all_contributing_variables = set()
    final_report = []

    for variable in variables:
        final_set, contributing_set, counteracting_set = calculate_firm_metrics(data, firm_name, variable, direction, threshold)
        firm_value = data.loc[data['Variables'] == variable, firm_name].values[0]
        industry_value = data.loc[data['Variables'] == variable, 'Industry Average'].values[0]

        if not counteracting_set.empty and 'Variable' in counteracting_set.columns:
            all_counteracting_variables.update(counteracting_set['Variable'].tolist())

        if not contributing_set.empty and 'Variable' in contributing_set.columns:
            all_contributing_variables.update(contributing_set['Variable'].tolist())

        final_report.append((variable, final_set))

    all_unique_variables = all_contributing_variables.union(all_counteracting_variables)
    all_unique_variables.add(selected_variable)

    filtered_counteracting_variables = filter_variables(list(all_counteracting_variables), fixed_list)

    missing_variables = set(get_missing_variables(filtered_counteracting_variables, filtered_dimensions))

    original_missing_variables = [var for var in fixed_list if var not in all_unique_variables]

```

```python
    missing_variables.update(original_missing_variables)
    missing_variables = list(missing_variables)

    filtered_report = []
    for variable, report in final_report:
        if variable not in missing_variables:
            filtered_report.append((variable, report))
        else:
            if variable in report['Variable'].values:
                continue
            filtered_report.append((variable, report))

    title = f"The factors on why the {selected_variable} of {firm_name} is {direction}\n"

    with open(output_file, 'w') as f:
        f.write(title)
        for variable, report in filtered_report:
            f.write(f"\nVariable Explained: {variable}\n")
            f.write(report.to_string(index=False))
            f.write("\n\n")

    # Print results to the terminal
    print(title)
    for variable, report in filtered_report:
        print(f"\nVariable Explained: {variable}\n")
        print(report.to_string(index=False))
        print("\n")

# Load the Excel file
file_path = r'C:\Users\jeroe\Code\Dit Werkt!\Nieuwe code\Data Firm Analysis revised.xlsx'
data = pd.read_excel(file_path)

# Generate report for the selected firm, dimension, and variable
generate_report(data, selected_firm, dimension, selected_variable, direction, threshold, output_file)
```

Prompt Builder Explanatory Query

```python
import openai
openai.api_key = key used in project is removed for privacy and fraud prevention reasons

# Abbreviations for the system message
abbreviations = """
ROA = Return On Assets
```

```
PM = Profit Margin
TA = Turnover Assets
OA = Operating Assets
PC = Production Cost
CA = Utilization rate of current assets
FA = Utilization rate of fixed assets
WIP = Work in progress
MC = Material cost
DMC = Distribution and marketing cost
FGS = Finished goods stock
DEB = Debtors
AC = Administrative Costs
RDC = Research and Development Costs
WLC = Works Labour Costs
WS = Work Sub-contracted
OPC = Other Production Costs
MS = Materials Stock
LB = Land and Buildings
PAM = Plant and Machinery
OFA = Other Fixed Assets
SPP = Sales/Production Policy
EX = Export Sales
MU = Metal Used
BC = Bought-out Components
"""

def read_report(file_path):
    with open(file_path, 'r') as file:
        content = file.read()
    return content

def explain_report(report):
    response = openai.ChatCompletion.create(
        model="gpt-4",
        messages=[
            {"role": "system", "content": "You are a helpful assistant. The following abbreviations  will be used:\n" + abbreviations},
            {"role": "user", "content": f"""Explain, without mentioning the table or measure of influence, why the variable is high or low by indicating the contributing and counteracting factors from the table.
            Note that the first variable is the selected variable and that all other tables when referring to contributing or counteracting refer to the influence on the selected variable.
            This is all the information that is required.
            This is the table:\n\n{report}\n\nExplanation:"""}
        ],
```

```python
        max_tokens=500
    )
    return response['choices'][0]['message']['content']

def process_report(file_path, output_file):
    report_content = read_report(file_path)
    reports = report_content.split("\n\nVariable Explained:")

    explanations = []
    for report in reports:
        if report.strip():  # Only process non-empty reports
            explanation = explain_report(report)
            explanations.append(explanation + "\n\n")

    with open(output_file, 'w') as file:
        file.writelines(explanations)

# File path to the saved report
report_file_path = 'final_report.txt'
output_file = 'explained_report.txt'

# Process the report and generate explanations
process_report(report_file_path, output_file)

```

Appendix E

Here are the example calculations to check the inputs for the ETG. The first example, is an example of a one level explanation by the explanation formalism algorithm, worked out manually. The parameters are firm 7, dimension = 1, variable = FA, the direction = high and threshold = 0.85. The direction is high since the FA of firm 7 is 0.441 and the industry average is 0.404.

- FA is calculated using LB + PAM + OFA. Using the equation in 3.1.1 it can is calculated that:
  - The measure of influence for LB = 0.311 + 0.184 + 0.005 - 0.404 = 0.096 and therefore contributing.
  - The measure of influence for PAM = 0.214 + 0.123 + 0.005 - 0.404 = -0.062
  - The measure of influence for OFA = 0.214 + 0.184 + 0.007 - 0.404 = 0.001
  - LB is the contributing parsimonious set since LB influence is 0.096/(0.096 + 0.001) = 0.99 which exceeds the threshold of 0.85. While OFA is contributing it's percentage of influence is 0.01 and therefore not part of the contributing parsimonious set.
  - PAM is the only counteracting variable which means it's influence is 1 and there the counteracting parsimonious set.

The resulting explanatory tables of the code are provided in Appendix E. By comparing the results it can be concluded that the code correctly calculates the measure of influence, identifies contributing and counteracting variables and creates the correct parsimonious sets.

```
The factors on why the FA of Firm 7 is high

Variable Explained: FA

  Firm Variable  Industry Average  Firm Value Measure of Influence Contributing/Counteracting
Firm 7       FA             0.404       0.441
Firm 7       LB             0.214       0.311                0.096               Contributing
Firm 7      PAM             0.184       0.123               -0.062              Counteracting
```

E1. Python Explanatory Table firm 6, dimension = 1, threshold =0.85, variable = FA

This is an example of a multi level explanation by the explanation formalism algorithm, worked out manually. The parameters are firm 2, dimension = 5, variable = ROA, direction is high and threshold = 0.85. For firm 2 the ROA is 0.239 and the industry average is 0.137, meaning the direction is set to high.

- ROA is calculated using PM * TA, using the equation in 3.1.1 it is calculated that:
  - The measure of influence for PM = 0.199 * 1.14 - 0.137 = 0.089 and therefore contributing.

  - The measure of influence for TA = 0.12 * 1.2 -0.137 = 0.007 and therefore contributing.
  - PM is the contributing parsimonious set since PM percentage of influence is 0.0899 / (0.0899 + 0.0.007) = 0.9278, which exceeds the threshold of 0.85.
  - Therefore TA does not appear in the analysis, PM does and is investigated further.

- PM is calculated using 1 - (AC + PC + RDC + DMC).
  - The measure of influence for AC = 1 - (0.04 + 0.722 + 0.007 + 0.072) - 0.12 = 0.039
  - The measure of influence for PC = 1 - (0.079 + 0.635 + 0.007 + 0.072) - 0.12 = 0.087
  - The measure of influence for RDC = 1 - (0.079 + 0.722 + 0.01 + 0.072) - 0.12 = -0.003
  - The measure of influence for DMC = 1 - (0.079 + 0.722 + 0.007 + 0.116) - 0.12 = -0.044
  - AC and PC are in the contributing parsimonious set since their respective individual percentile influences are 0.31 (0.039/0.126) and 0.69 (0.087/0.126) which both don't exceed 0.85. However, together they form 1 which exceeds 0.85.
  - DMC is in the counteracting parsimonious set since it's individual influence is 0.94 (-0.044/-0.047).
  - Therefor RDC does not appear in the analysis, DMC is not investigated further since it is a counteracting variable. AC is in the analysis, but is not further investigated since it does not have a formula. PC is in the analysis and is investigated further.

- PC is calculated using MC + WLC + WS + OPC. Since a high PC means a lower PM and a lower ROA, the measure of influence is not multiplied by 1 but by -1 when the direction is high.
  - The measure of influence for MC = $(0.287 + 0.234 + 0.027 + 0.127 - 0.722) * -1 =$ 0.047
  - The measure of influence for WLC = $(0.221 + 0.333 + 0.027 + 0.127 - 0.722) * -1 =$ 0.014
  - The measure of influence for WS = not calculated since the value of WS is 0 and therefore has no influence on PC.
  - The measure of influence for OPC = $(0.127 + 0.333 + 0.234 + 0.027 - 0.722) * -1 =$ 0.001
  - All calculated variables are contributing but only WLC and MC are in the parsimonious set of contributing variables. This is because the individual percentage of

influence are 0.769, 0.229 and 0.002. This shows that WLC and MC is the smallest combination which exceeds 0.85.

– All variables are shown in the analysis, but only MC is further investigated since it is the only variable in the contributing parsimonious set which has a formula

– MC is determined using MC = 'MU': 'pos', 'BC': 'pos', meaning when MU or BC is higher than the industry average the MC is also high. Normally this means that MU or BC are contributing when higher than the industry average. However, since a high MC means a high PC means a low PM means a low ROA for this analysis a low MU or BC is a contributing variable.

– The industry average for MU is 1.440 and the firm value for firm 2 is 1.000 since the industry average is higher this means that MU is a contributing variable for a high ROA.

– The industry average for BC is 0.380 and the firm value is 0.390. The firm value is higher and therefore BC is a counteracting variable.

The resulting explanatory tables of the code are depicted in the figure Appendix E. First all measure of influences are compared and examined if calculated correctly, this holds. Furthermore, the code also identifies correctly the same contributing and counteracting variables. Notably, TA and its component variables OA, CA, FA etc are not included in the report since TA is not part of the contributing parsimonious set, RDC is not included since it is not part of the counteracting parsimonious set. WS is not included since its value is zero, OPC is not included since its not part of the contributing parsimonious set. The only notable thing which is different is the measure of influence for the qualitative variable MC. In the manual calculation this is not calculated since for MU does not take value from real numbers.

The factors on why the ROA of Firm 2 is high

Variable Explained: ROA

Firm	Variable	Industry Average	Firm Value	Measure of Influence	Contributing/Counteracting
Firm 2	ROA	0.137	0.239		
Firm 2	PM	0.120	0.199	0.0899	Contributing

Variable Explained: PM

Firm	Variable	Industry Average	Firm Value	Measure of Influence	Contributing/Counteracting
Firm 2	PM	0.120	0.199		
Firm 2	AC	0.079	0.040	0.039	Contributing
Firm 2	PC	0.722	0.635	0.087	Contributing
Firm 2	DMC	0.072	0.116	-0.044	Counteracting

Variable Explained: PC

Firm	Variable	Industry Average	Firm Value	Measure of Influence	Contributing/Counteracting
Firm 2	PC	0.722	0.635		
Firm 2	MC	0.333	0.287	0.047	Contributing
Firm 2	WLC	0.234	0.221	0.014	Contributing

Variable Explained: MC

Firm	Variable	Industry Average	Firm Value	Measure of Influence	Contributing/Counteracting
Firm 2	MC	0.333	0.287		
Firm 2	MU	1.440	1.000	0.44	Contributing
Firm 2	BC	0.380	0.390	-0.01	Counteracting

E2. Python Explanatory Table Firm 2, dimension = 5, t=0.85, variable = ROA

Appendix F

Use of AI

Following both the policy of Tilburg University and Turku University in this thesis it should be made clear when, where and how any AI tools are being used. The following AI tools have been used in this thesis: Consensus, ChatGPT, Elicit and Grammarly. The outcomes of these tools have been assessed and evaluated before being used.

Academic Papers
For finding academic papers, alongside traditional tools such as WorldCat Discovery, Google Scholar, and Scopus, newer tools like (Consensus, 2024) and (Elicit, 2024) have been utilised. Both Consensus and Elicit offer summaries of the located academic papers. However, their summarization features have been found to be inaccurate, resulting in their primary use being limited to discovering new papers.

Brainstorm Terms
ChatGPT has been employed to explore and generate new terminology. In instances where certain parts of this thesis lacked established terms, ChatGPT was utilised to brainstorm appropriate terminology. For example, the term 'explanatory text generator' was created for this thesis artefact. This process was carried out using the GPT-4 model (OpenAI, 2024).

Formatting Overleaf
ChatGPT has been utilised to insert figures, formulas, and tables. All formulas were copied and processed by ChatGPT to be converted into Overleaf format. Additionally, data was provided to create tables; for instance, the layout of Table 3 was generated by ChatGPT. This process employed a GPT model designed for writing in Overleaf (OverleafGPT, 2024).

Improving Text
ChatGPT has been utilised to advise on and enhance certain wording and sentence structures, formatting them to better align with academic standards. No new text has been generated by ChatGPT; instead, the original text was submitted to ChatGPT with the consistent prompt: 'Please advise on how to rephrase the following text to make it more suited for an academic paper using British English Grammar: [given text].' This approach ensures uniformity in the rephrasing process, utilising the standard GPT-4 model (OpenAI, 2024).

Furthermore, Grammarly, in combination with Overleaf, has been employed to enhance grammar, clarity, conciseness, and tone through its proofreading capabilities (Overleaf, 2024).

Randomisation
ChatGPT has been used for randomisation. For example the test scenario's for the explanatory

query inputs have been randomised by ChatGPT. This process was carried out using the GPT-4 model (OpenAI, 2024).

Writing Code

ChatGPT has been utilised to suggest and optimise portions of the code. For instance, new code was developed for the module responsible for transforming the OpenAI output format to HTML. Additionally, if the optimised code demonstrated improvements in speed or readability over the original, it was adopted. This process employed a GPT made for writing Python code (PythongGPT, 2024).